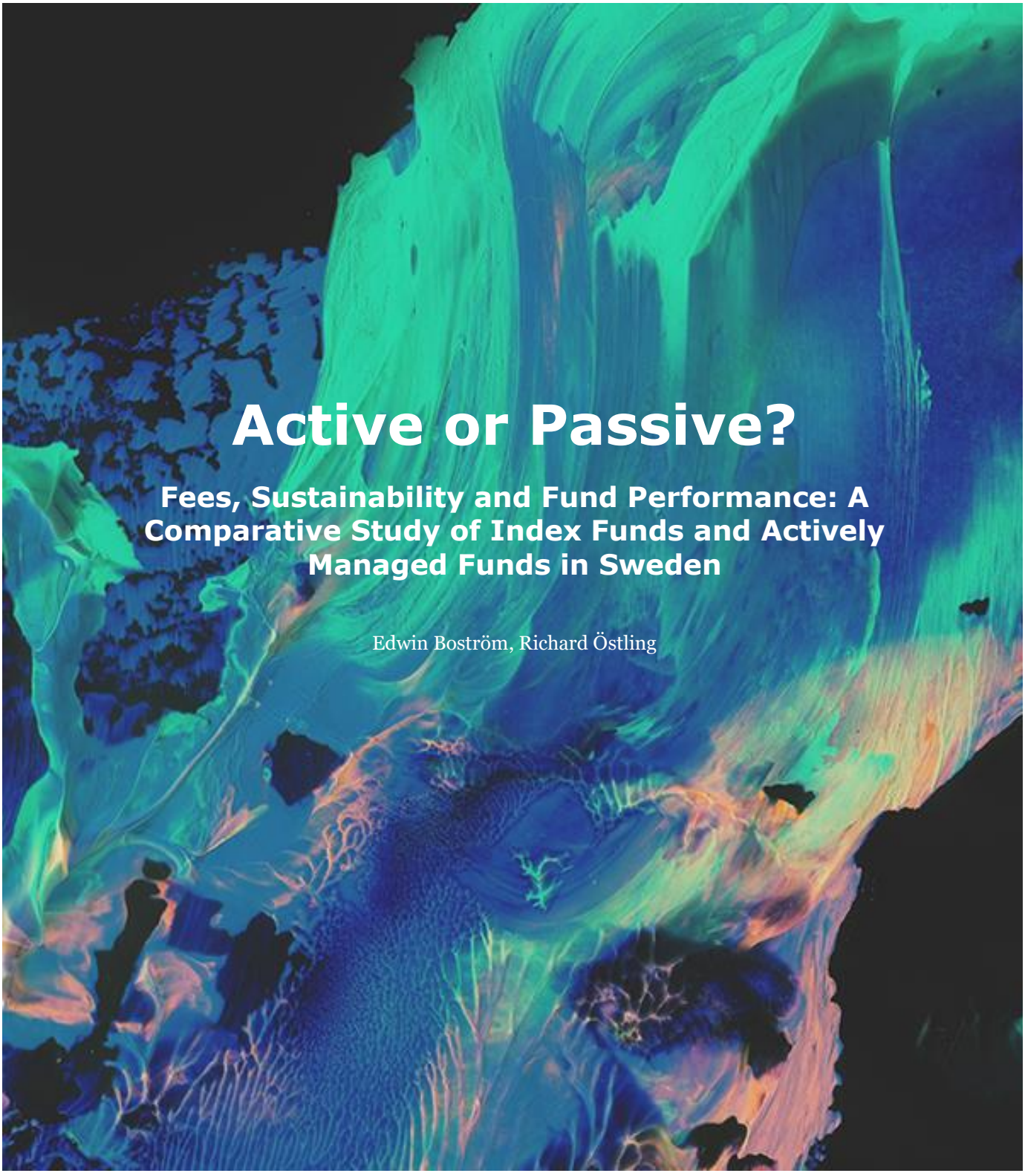
UMEÅ SCHOOL OF BUSINESS,
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Active or Passive?

Fees, Sustainability and Fund Performance: A Comparative Study of Index Funds and Actively Managed Funds in Sweden

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Abstract

This thesis examines whether fees and sustainability grades are associated with fund returns in Sweden when controlling for standardised fund risk level, and how fund performance differs between index funds and actively managed funds. The study focuses on the Swedish fund market, where fund ownership is widespread and sustainability has become an increasingly important factor in investment decisions. By combining traditional fund characteristics, such as fees and risk, with sustainability grades, the thesis aims to contribute to the discussion on what factors are associated with investor-relevant fund performance.

The theoretical framework is based on the Efficient Market Hypothesis, Modern Portfolio Theory, ESG theory. These theories are used to interpret the relationship between fund type, fees, sustainability grades, risk level and fund returns. The empirical analysis is based on a quantitative research design using monthly observations from 84 Swedish funds over a five-year period. The sample includes both index funds and actively managed funds. The main dependent variable is net return, while gross return is used as an alternative measure. The main empirical analysis applies random-effects panel regressions with standard errors clustered at the fund level. A pooled OLS model is also estimated as a baseline comparison, while additional specifications using gross returns and interaction terms are used to assess whether the findings are robust across alternative model formulations.

The results show that fund type is associated with fund performance. In the main model specifications, actively managed funds are associated with lower returns compared to index funds. This is consistent with previous research suggesting that active funds often struggle to outperform passive alternatives after costs. The findings also show that sustainability grades are statistically associated with fund returns. Since the Morningstar sustainability score is measured as a sustainability risk score, the negative ESG coefficient indicates that funds with lower sustainability risk are associated with higher returns in the sample. Fees are also associated with returns, although the relationship is more complex than expected. The positive fee coefficient suggests that fees may capture broader fund characteristics, such as fund type, investment strategy or market exposure, rather than only the cost paid by investors. The standardised risk level does not show a statistically significant relationship with monthly fund returns.

Overall, the thesis finds that fees and sustainability grades are associated with fund returns in the Swedish sample, while standardised risk level is not statistically significant. The results also indicate that index funds performed better than actively managed funds during the analysed period. The findings should be interpreted as statistical associations rather than causal effects.

Keywords: fund returns, fees, risk, sustainability grades, ESG, index funds, actively managed funds.

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1. Introduction

1.1 Problem background

“Investing is a funny business. It’s really easy to be average. Just buy an index fund.”

- Howard Marks, Co-Founder of Oaktree Capital Management.

The first Swedish funds were created in 1958 by Ragnar and Gösta Åhlén, Sweden's total fund capital amounted to 136 million crowns one decade later (Swedish Investment Fund Association, n.d.). Sweden implemented tax reliefs related to fund ownership, which allowed 20% of each crown placed in a fund to be deducted from the person's income taxes, this tax reform made fund ownership nearly six times more popular between 1979 and 1982. In the mid 1980s, “Allemansspar” was introduced, which offered fund ownership as an option in a traditional savings account, furthermore, fund returns were completely tax-free at this time. During the 1990s, the tax subsidies were redacted, however harmonising EU regulations, fund ownership as a part of the pension plan and the premium pension allowed the funds’ growth in popularity to continue. In 2012, Sweden implemented the “Investeringssparkonto” (ISK), which is a type of savings account for capital investments; there are no taxes on capital gains under ISK regulations (Swedish Investment Fund Association, n.d.). Today, funds are more popular in Sweden than in any other country in the world, 70% of Swedish citizens have fund ownership by their own decision, that is, excluding pension savings (Swedish Investment Fund Association, n.d.).

There are roughly 3000 Swedish funds available to investors (Swedish Investment Fund Association 2024a, p. 14), these are generally divided into four categories; equity funds, mixed funds, interest funds and hedge funds, however each category has several sub-categories.

Funds may also be categorised based on their management; active fund management and passive fund management. A typical form of passive management are index funds (Vanguard, n.d.). Index funds aspire to match their underlying market or index as closely as possible. They do this by buying a representative part of the stocks or bonds that constitute the index. The risk profile of index funds is directly correlated to the risk profile of the underlying market it wishes to track, thus, index funds generally reduce manager-specific risk by tracking a broad market index. Index funds usually offer low management fees as there is limited active management needed for fund operations (Vanguard, n.d.).

While index funds try to match their underlying index, actively managed funds try to outperform their underlying index (Vanguard, n.d.). They do this by utilizing the fund managers’ expertise in hand-selecting stocks or bonds that the fund should invest in. Thus, the risk profile consists of both the underlying market and the risk that the fund manager underperforms the underlying market because of poor investment decisions. Actively managed funds generally have higher fees than passively traded funds because a lot of management is needed to outperform the market (Vanguard, n.d.).

The fund fee is the percentage that an investor is charged to claim ownership stakes in a fund (Servaes & Sigurdsson, 2022, p. 1). The total fee often consists of several different fees. One of these is the management fee, which is what the fund managers charge for

their ongoing commitment in the fund operations. Another common fee is the performance fee, which is gaining increasing popularity among mutual funds. Additionally, there are two types of performance fees, a symmetric performance fee rewards fund management companies for outperforming their underlying index, but induce an equal penalty for underperformance. An asymmetric performance fee rewards fund management companies for outperformance but induces no penalty for underperformance (Servaes & Sigurdsson, 2022, p. 1). A Swedish index fund typically has a total fee of 0,2%, and a global index fund typically has a total fee of 0,4% (Finansinspektionen, 2026). Whereas an actively managed Swedish fund typically has a total fee of 1,3%, and the same for global actively managed funds (Finansinspektionen, 2026). This study aims to examine whether the large difference in total fee is justified or not.

A higher risk should generate a higher return, and vice versa, this relationship is widely accepted and integrated in financial literature and highlighted in the CAPM formula (Sharpe. W, 1964).

The relationship may be viewed as a trade-off, if the potential return is higher, the likelihood of achieving that return is reduced (Beckham, H, W. 2019, p. 1). Finance literature often divides investment risks into two categories; systematic risk and unsystematic risk. Unsystematic risks are risks that affect individual securities such as legal risk, competition and operational risks. The unsystematic risk is generally high if a portfolio consists of only a few securities, but can be practically eliminated if the portfolio is more diversified. Systematic risks are risks that affect virtually all companies, such as interest rate risks, inflation risks and global political conflicts. Systematic risk cannot be reduced by diversification, but is a direct risk to investments in any security, thus, systematic risk is the only risk investors are rewarded for bearing with higher potential returns (Beckham, H, W. 2019. p. 1-2). Since funds generally hold a large number of securities in its portfolios, the unsystematic risk is close to zero.

While financial theory often conceptualises risk through market-based measures such as beta or return volatility, this study uses a standardised fund risk category as a practical control variable. The purpose is therefore not to estimate systematic risk in the strict CAPM sense, but to account for broad differences in the risk profile communicated to retail investors.

Sustainability in investing has quickly shifted from being a side-issue to a central part of investment strategy (UN Global Compact, 2025). Many investors believe that ESG integration reduces portfolio volatility and a vast majority of investors plan to increase their sustainable investments in the near term (UN Global Compact, 2025). The shift in investor strategy has resulted in a large number of sustainable funds (SIFA, n.d). These funds work to ensure that portfolio companies are sustainable and report details on their ESG work. Furthermore, a large number of funds have shifted from excluding certain industries such as arms, tobacco or gambling to only investing in positive companies that are at the forefront of the sustainability movement (SIFA, n.d). Today, all funds are graded based on their ESG performance by a number of ratings agencies, one of which is Morningstar, which rates funds between 0-100 on their aggregated performance in environmental, social and governance across their portfolio (Avanza, 2026). A lower Morningstar sustainability risk score, indicates lower sustainability risk and therefore a stronger sustainability profile. Thus, investors do not need to choose between pure

sustainability or no sustainability, but can allocate their capital across the sustainability spectrum. Strong ESG performance has been linked to lower idiosyncratic and extreme risks in individual stocks (Xu et al., 2025, p. 1). It has also been shown to decrease the levels of corporate earnings management, improve firm reputation and decrease stock volatility (Xu et al., 2025, p. 1). Therefore, both risk and sustainability are vital factors for this study.

1.2 Problematization

Funds are popular among investors and are often part of a person's savings. Funds are thus also extensively studied in the field of finance. Despite the extensive literature on fund performance, several important gaps remain. There are previous studies on the comparison of the fund return between index funds and actively managed funds, such as that of Fortin & Michelson (1999). These researchers compared the net returns of the two types of funds over a 15 year period and found statistical evidence that index funds outperform actively managed funds (Fortin & Michelson, 1999, p. 78). The study is relatively old and may be outdated, the time period for the study was 1981-1995. Furthermore, the sample consisted of only US-based funds, it can therefore not be said with certainty that the relationship also exists in Swedish funds. The researchers also highlight that their chosen time period created a lack of index funds and proposed future research to examine the same relationship once more index fund data is available (Fortin & Michelson, 1999, p. 78). The study did not use any control variables other than the total fee, failing to capture the potential influence from risk level and sustainability, thus, the study can be reproduced and improved in a more modern context.

A study conducted by Knutsson & Edlund investigated the same matter but in a Swedish context with data from 2011-2021. The study found similar results to that of Fortin & Michelson (1999). Due to a small sample of only seven actively managed funds and one index fund, the results were rendered statistically insignificant (Knutsson & Edlund 2025, p. 50). This study did not use any control variables other than the total fee.

Previous research on sustainability-grades and fund returns is quite diffuse. A study conducted by Piller & Harryzon (2020) investigated the relationship in the Nordics between 2015-2020 and found no statistically significant relationship between sustainability grades and fund performance (Piller & Harryzon, 2020, p. 28). Another study conducted by Fornander & Taipale (2024) investigated the relationship in Swedish funds during a 3-year time perspective, controlled for total fund assets and fund age, found a positive relationship between sustainability grades and fund performance (Fornander & Taipale, 2024, p. 40). Furthermore, a study conducted by Sjöström & Venäläinen (2018) found a statistically significant negative relationship between ESG factors and stock returns on the Swedish stock market between 2013-2017 (Sjöström & Venäläinen, 2018, p. 27), whereas individual stock returns are not the same as fund returns, it is an interesting find in a closely related area.

Several previous studies on sustainability grades and fund performance use different sample sizes, time horizons and periods and control variables but have one shared limitation; they choose to regard funds as one homogenous group. They also make no mention of risk levels in the funds, which reduces depth since risk and sustainability are directly interlinked. Thus, there is an apparent need for a more exhaustive study on how these factors are associated with actively managed funds and index funds in a Swedish

setting. There is also a need to analyse sustainability grades and risk proxy together with fund type and fees, rather than treating these factors in isolation.

This study aims to bridge the gap between two extensively studied areas in finance; index funds versus actively managed funds, and the relationship between sustainability grades and fund performance. To add more depth to the analysis, risk level is included as a control variable to account for differences in the general risk profile of the funds. The variable is not intended to provide a precise market-based risk estimate, but rather to capture the standardised risk category communicated to investors.

1.3 Research purpose

The purpose of this study is to examine how fees and sustainability grades are associated with fund performance in Sweden, while controlling for differences in fund risk level. The study also examines how the association may differ between actively managed funds and index funds.

The study aims to contribute to the understanding of sustainability as a relatively new criterion in financial analysis and its association with the widely used mutual funds. Insights may provide investors with a new perspective when selecting funds for their portfolios.

Furthermore, fund management companies may use the findings to reassess their strategies regarding sustainability integration, risk levels and fee structures.

1.4 Research question

This thesis addresses the following main research question:

How do the associations between fees, sustainability grades, and fund returns in Sweden differ between actively managed funds and index funds, after controlling for standardised fund risk level?

1.5 Theoretical contribution

This thesis aims to contribute to research on fund performance. The theoretical framework is used to interpret the empirical results in relation to previous research. The theoretical framework will also be used to highlight the significance of the empirical results, which may in turn lead to future research opportunities within the field.

Significant statistical results would highlight the impact of sustainability grades and risk level on net fund returns, as well as the differences of the relationship between index funds and actively managed funds. This would contribute to the existing literature within the field of fund returns, sustainable finance and risk and return. Significant statistical results may contribute to investment decision-making by clarifying whether fees, sustainability grades and risk levels are associated with fund returns. The findings may therefore provide investors with additional information when comparing index funds and actively managed funds.

Non-significant statistical results may be viewed as a result in itself, and will still contribute to the understanding of sustainability grades and risk levels in the fund

universe. It would open up for more research in the space of index funds and actively managed funds, as well as sustainability grades and risk levels as drivers of fund returns.

1.6 Practical contributions

The study will explore the relationship between fund fees, sustainability grades, risk level and fund returns and examine how the relationship differs between index funds and actively managed funds. The findings may be useful for investors who consider fund fees, sustainability grades and risk levels when selecting funds. By comparing index funds and actively managed funds, the study may also provide insight into whether higher fees are associated with higher or lower investor-relevant returns.

The findings may also be relevant for fund management companies. After reviewing the results, fund management companies may wish to increase their sustainability grades and review their risk level and fee structure, depending on what the results are.

1.7 Delimitations

This study is subject to several delimitations that define its scope and clarify what is intentionally excluded. These choices have been made to ensure feasibility at an advanced academic level while maintaining a clear focus on the research question: *How do the associations between fees, sustainability grades, and fund returns in Sweden differ between actively managed funds and index funds, after controlling for standardised fund risk level?*

First, the study is geographically limited to the Swedish fund market. This implies that only funds active in the Swedish market and primarily regulated within the Swedish financial system, are included. This improves comparability between funds, since they operate in a similar investment environment. At the same time, it limits the possibility of generalising the results to other markets. For instance, the widespread use of the ISK and the high rate of fund ownership are specific characteristics of the Swedish context, which may influence both fund performance and investment decisions.

Second, the study focuses exclusively on mutual funds and excludes other financial instruments such as exchange-traded funds (ETFs), individual stocks, derivatives and alternative investments, including private equity and real estate funds. Although ETFs share similarities with index funds, they are traded differently and often involve distinct fee structures, which could introduce additional complexity. By restricting the analysis to traditional mutual funds, comparability between actively managed funds and passively managed (index) funds is maintained.

Third, funds are categorised into two primary groups: index funds and actively managed funds. This classification simplifies the analysis but inevitably overlooks variations within each category. Certain funds apply semi-active strategies or enhanced indexing approaches, thereby blurring the distinction between passive and active management. Such hybrid strategies are not explicitly accounted for in this study.

Fourth, the analysis is restricted to a selection of key independent variables: fees, sustainability grades. While these variables are central to the research question, fund returns are influenced by a broader set of factors, including macroeconomic conditions,

sector allocation, currency exposure, fund size and managerial expertise. These variables are not incorporated into the primary models, implying that the results should be interpreted as partial explanations rather than comprehensive determinants of fund performance. The study does not include fund size, fund age, sector allocation due to data limitations. These variables may influence fund performance and should therefore be considered in future research.

Fifth, sustainability is measured using ESG scores provided by Morningstar, a widely used rating provider within the Swedish fund market. The ESG score is expressed on a scale from 0 to 100, where a lower score indicates lower sustainability risk and therefore a stronger sustainability profile. This introduces a limitation related to measurement consistency, as ESG ratings may vary across providers due to differences in methodology and underlying criteria. Consequently, the sustainability variable reflects the perspective of the selected provider rather than an absolute measure of environmental, social and governance performance.

Sixth, the study relies on historical data over a specified time period (April 2021 to April 2026), implying that the results are inherently backward-looking. The relationship between fees, sustainability, risk and returns may change over time as a result of evolving market conditions, regulatory developments or shifts in investor preferences. Therefore, the findings should be interpreted with caution when considering future fund performance.

Finally, the study does not account for individual investor behaviour, such as timing of investments, variations in risk preferences or specific tax considerations beyond the general Swedish framework. Instead, a simplified assumption is applied, whereby investors hold funds over the analysed period. While necessary for quantitative analysis, this abstraction may differ from actual investment behaviour. Also, the study does not intend to remove share classes belonging to the same underlying fund. This may be included multiple times in the dataset. This may increase the similarity between some observations and potentially overweight certain fund strategies. This is acknowledged as a potential limitation of the dataset later on.

In summary, these delimitations enable a focused and structured analysis of the Swedish fund market, while also defining its boundaries. Acknowledging these limitations is essential for an accurate interpretation of the results and for identifying potential areas for future research.

1.8 Choice of subject and preconceptions

The choice of subject for this study is grounded in both the importance of fund investing in Sweden and the growing relevance of sustainability considerations in financial decision-making. As previously outlined, a substantial share of Swedish citizens participate in fund markets, making funds a central component of private wealth accumulation. At the same time, investors are increasingly required to consider additional dimensions beyond financial returns, including sustainability grades and risk exposure. This creates a relevant context for analysing how traditional financial factors, such as fees and risk, interact with sustainability measures in explaining fund performance.

Another motivation for the selected topic is the ongoing discussion regarding the relative performance of active and passive fund management. Index funds have gained significant popularity due to their low fees and their ability to closely track their underlying market (Cremers et al., 2016; Investment Company Institute, 2025). In contrast, actively managed funds aim to outperform their benchmark through selective investment strategies, which is commonly used to justify higher management fees. This study contributes to this discussion within a Swedish context by empirically comparing these two categories, while also incorporating additional variables that may be associated with fund returns .

In terms of preconceptions, the study needs to acknowledge that certain expectations were present at the outset of the study, influenced by existing research and general market perspectives. One initial preconception was that some actively managed funds may justify higher fees through superior management and higher gross returns. Previous research suggests that higher fund fees may reduce investors' net returns unless these costs are offset by superior performance (Servaes & Sigursson, 2022; Malkiel 2015). This study therefore treats the relationship between fees and performance as an empirical question. Furthermore, there is another view that actively managed funds, on average, may struggle to consistently outperform their underlying market, especially after accounting for fees.

Regarding sustainability, prior research presents mixed findings, which motivates a more neutral initial position. There is, however, an awareness of arguments suggesting both positive and negative relationships between ESG performance and financial returns. As a result, no definitive assumption is made concerning the direction of this relationship.

To reduce the influence of the preconceptions, the study applies a quantitative and empirical approach, relying on statistical analysis rather than subjective interpretation. The intention is to allow the data to determine the relationships between the selected variables. Nevertheless, full objectivity cannot be entirely ensured, as methodological choices, including model specification, variable selection and data sources, may still reflect underlying assumptions.

By explicitly acknowledging these preconceptions, the study aims to increase transparency which strengthens the credibility of its findings.

2. Theoretical and Conceptual Framework

This chapter presents the theoretical framework that underpins the study. It begins by outlining key economic and financial theories relevant to the research, establishing a conceptual foundation for the analysis.

2.1 Theory

2.1.1 Efficient Market Hypothesis

The Efficient Market Hypothesis, developed by Eugene Fama, proposes that financial markets are informationally efficient, implying that asset prices fully and instantaneously reflect all available information (Fama, 1970, p. 383-384). As a result, investors are not expected to consistently achieve abnormal returns, defined as returns exceeding the market average on a risk-adjusted basis, through stock selection or market timing.

Efficient Market Hypothesis is commonly divided into three forms: weak, semi-strong and strong efficiency (Fama, 1970, p. 383-388). The semi-strong form, which is particularly relevant in the context of mutual funds, assumes that all publicly available information is already incorporated into asset prices. Under this assumption, attempts by active fund managers to outperform the market using publicly available data should, on average, be unsuccessful (Malkiel, 1995).

For this thesis, this means that any return differences between active and index funds should be interpreted carefully, especially after fees and risk are considered. Consequently, after accounting for costs such as management fees, actively managed funds are expected to underperform passive strategies that replicate the market index (SPIVA, n.d.; Malkiel, 1995, p. 550-552; Fama and French, 2010, p. 1915-1917).

Efficient Market Hypothesis has been subject to extensive criticism. Malkiel (2003, p. 4-5) explains that critics argue that stock prices may be partially predictable due to behavioural factors and recurring return patterns. In such contexts, skilled managers may be able to exploit temporary inefficiencies and generate excess returns. This introduces the possibility that active management may outperform passive strategies, particularly in markets characterised by lower transparency, limited analyst coverage or behavioural distortions (Cremers and Petajisto, 2009, p. 3330-3332). This theory is relevant to the comparison between index funds and actively managed funds, since active management depends on the possibility of identifying mispriced assets or timing the market.

Another explanation for why actively managed funds often struggle to outperform index funds is the phenomenon commonly referred to as “closet indexing”. According to Cremers and Petajisto (2009, p. 3330-3331), many actively managed funds maintain portfolios that closely resemble their benchmark indices despite charging substantially higher management fees. Funds with low active share are therefore often exposed to similar market movements as index funds while still imposing higher costs on investors.

The implications of EMH are closely related to investment costs. If markets are largely efficient and persistent excess returns are difficult to achieve, management fees, transaction costs and other expenses become particularly important for investor

outcomes. These costs directly reduce net returns and may therefore create a structural disadvantage for actively managed funds compared with lower-cost index funds (Malkiel, 1995, p. 550-552; Carhart, 1997, p. 67-68). This conclusion depends on the assumption that active managers do not possess persistent skill.

This creates a structural challenge for active fund managers. In order to justify higher fees, actively managed funds must generate sufficiently differentiated investment exposure and consistently outperform the market after costs (Cremers and Petajisto, 2009, p. 3332-3333). But if active managers remain too close to their benchmark portfolios, the potential for outperformance becomes limited while investors still bear the additional fee burden.

2.1.2 Modern Portfolio Theory

The Modern Portfolio Theory (MPT), introduced by Harry Markowitz, provides a formal theory for analysing investment decisions under uncertainty (Markowitz, 1952, p. 77-79). The theory assumes that investors are rational and risk-averse, meaning that they seek to maximise expected returns for a given level of risk.

A central concept within MPT is diversification, which enables investors to reduce unsystematic risk by holding a combination of assets whose returns are not perfectly correlated (Markowitz, 1952, p. 77-91). Within this theory, higher expected returns are interpreted as compensation for higher levels of systematic risk, which cannot be eliminated through diversification (Markowitz, 1952, p. 77-79). Consequently, differences in portfolio performance are expected to be closely related to differences in risk exposure.

Despite its theoretical significance, MPT relies on several simplifying assumptions, including normally distributed returns, stable correlations and fully rational investor behaviour (Markowitz, 1952). In practice, these assumptions may not hold. Market frictions and behavioural biases may lead to deviations from the efficient frontier (Malkiel, 2003, p. 7-9; Kahneman & Tversky, 1979, p. 263-268), implying that the relationship between risk and return may be more complex than predicted by the theory.

2.1.3 Sustainable Finance and ESG

The integration of environmental, social and governance (ESG) factors into investment decisions is grounded in the broader field of sustainable finance. Traditional financial theory assumes that investors primarily consider risk and return (Markowitz, 1952, p. 77-79; Sharpe, 1964, p. 425-426). ESG extends this framework by incorporating non-financial considerations and long-term value creation (Friede et al., 2015, p. 210-213).

One perspective within ESG theory suggests that firms with strong sustainability practices are better positioned to manage risks related to regulation, environmental impact and stakeholder relations (Friede et al., 2015, p. 210-213). This may result in improved operational performance, lower cost of capital and increased resilience during periods of economic instability. From this standpoint, investments with higher sustainability ratings may achieve greater risk-adjusted returns over the long term (Friede et al., 2015, p. 210-213).

An alternative perspective highlights potential trade-offs associated with ESG investing. By restricting the investment universe based on sustainability criteria, diversification opportunities may be reduced and potentially profitable firms or industries may be excluded. Hong and Kacperczyk (2009, p. 15-17) show that controversial stocks may earn higher expected returns because some investors avoid them, implying that exclusion-based investment strategies may involve an opportunity cost.

The relationship between ESG and financial performance is therefore theoretically ambiguous. Investor preferences may also affect expected returns. If a significant share of investors favours sustainable assets, increased demand may raise prices and reduce expected future returns. However, less sustainable assets may offer higher expected returns as compensation for lower demand or weaker investor preferences (Pástor et.al., 2021, p. 550-552; Hong and Kacperczyk, 2009, p. 15-17).

Consequently, the empirical relationship between ESG and financial performance remains an open question within the literature.

2.2 Connecting Thesis to Theory

The theories above are used to interpret the variables included in the study. Efficient Market Hypothesis is used to discuss whether actively managed funds perform differently from index funds. MPT motivates the inclusion of risk level, while cost efficiency theory explains why fees are expected to matter for net returns. ESG theory is used because the relationship between sustainability and fund returns is not clear in previous research.

The Efficient Market Hypothesis serves as a central reference point for evaluating differences in performance between index funds and actively managed funds. If the results indicate that actively managed funds do not outperform passive alternatives after accounting for fees, this would support the notion of informationally efficient markets, where excess returns are difficult to achieve systematically (Fama, 1970, p. 383-388; Malkiel, 2003, p. 31-33). On the other hand, evidence of persistent outperformance among actively managed funds would suggest the presence of market inefficiencies or managerial skill, thereby challenging the assumptions of market efficiency.

The Modern Portfolio Theory provides the foundation for interpreting the relationship between risk and return. A positive association between risk levels and fund returns would be consistent with the theoretical prediction that higher returns compensate for higher risk exposure (Markowitz, 1952, p. 77-79; Sharpe, 1964, p. 425-426). On the other hand, if this relationship is weak, absent or negative, it may indicate deviations from theoretical assumptions, potentially reflecting market frictions, imperfect diversification (Malkiel, 2003, p. 7-9; Kahneman and Tversky, 1979, p. 263-268).

The interpretation of sustainability grades is informed by competing perspectives within ESG theory. A positive relationship between sustainability and returns would support the argument that ESG integration contributes to long-term performance through improved risk management and firm quality (Friede et al., 2015, p. 210-213). A negative relationship would instead be consistent with the view that sustainability constraints limit diversification or exclude high-performing assets (Hong and Kacperczyk, 2009, p. 15-17; Pástor et al., 2021, p. 550-552). An insignificant

relationship may indicate that ESG characteristics are already reflected in market prices or that their financial implications are context-dependent.

Together, these theoretical perspectives guide the analysis. The purpose is not to test the theories in isolation, but to use them as interpretative tools for understanding whether the empirical results support, challenge or add nuance to previous research on fund performance.

2.3 Conceptual Framework

The conceptual framework shows how the main variables in the study are connected. The dependent variables are net return and gross return. The main independent variables are fund type, fees and ESG score, while risk level is included as a control variable. The framework also reflects the comparison between index funds and actively managed funds.

2.3.1 Net Return

Net return is included as a dependent variable within the conceptual framework. Net return represents the financial performance of a fund after all relevant fees have been deducted and therefore reflects the actual return received by investors over a specified period. Similar to fund return, net return serves as an outcome variable that the study seeks to explain through the independent variables. However, while fund return captures performance in broader terms, net return explicitly incorporates the effect of costs, making it a more precise measure of investor-relevant performance.

2.3.2 Gross Return

Gross return is included as an alternative dependent variable. The gross return represents the financial performance of a fund over a specified period before deducting any costs.

2.3.3 Fees

Fees are expected to have a direct negative relationship with fund returns, as they represent a cost to the investor that reduces net performance. This relationship is particularly relevant when comparing index funds and actively managed funds, as the latter generally involve higher fees. The conceptual framework therefore assumes that, all else equal, higher fees will be associated with lower net returns.

2.3.4 Risk level

Risk level is included as a control variable to account for differences in the general risk profile of the funds. The variable is based on the standardised fund risk category reported to investors on a scale from 1 to 7. This type of indicator is designed to rank funds according to increasing risk, primarily based on volatility, and is used in European fund documentation (European Commission, 2017).

In this thesis, risk level is therefore interpreted as an investor-facing risk proxy rather than a precise econometric measure of risk. The expected relationship between risk level and return is informed by Modern Portfolio Theory, which suggests that higher risk

should be associated with higher expected return. However, because the variable is categorical and standardised, the results are interpreted cautiously.

2.3.5 Sustainability scores (ESG)

ESG are included as an additional independent variable reflecting the increasing importance of non-financial factors in investment decisions. The expected relationship between ESG and returns is theoretically ambiguous. On the one hand, higher sustainability may lead to better long-term performance through improved risk management and governance. On the other hand, it may constrain investment choices and potentially reduce returns. Therefore, no fixed direction is assumed for ESG before the empirical analysis.

2.3.6 Fund type

The thesis incorporates fund type (index versus actively managed) as a categorical dimension. This distinction is not only central to the research question but also influences how the independent variables interact with fund returns. For example, fees are structurally lower in index funds, while risk and sustainability profiles may differ between the two categories. The comparison between these fund types allows the study to assess whether fund performance differs by management strategy, and whether the ESG-return relationship differs between index funds and actively managed funds.

Although other factors may also affect returns, they are excluded in line with the delimitations of the study to maintain analytical clarity and due to data limitations. By structuring the analysis in this way, the conceptual framework provides a clear link between theory, variables, and empirical methodology. Thereby, the study remains focused on the key relationships of interest while allowing for systematic testing and interpretation of the results.

2.4 Literature Review

2.4.1 Fees and Fund Performance

The relationship between fees and fund performance has been widely examined in financial literature, particularly in the context of comparing actively managed funds and index funds. According to Fama and French (2010, p. 1915-1917), there is limited evidence that active fund managers are able to consistently outperform the market after accounting for fees. Their findings suggest that observed outperformance is largely attributable to luck rather than skill. Similarly, Malkiel (1995, p. 550-552) demonstrates that actively managed mutual funds tend to underperform passive index funds over longer time horizons, particularly once management fees are deducted.

Further empirical support is provided by reports from S&Ps, which consistently show that a majority of actively managed funds fail to outperform their benchmark indices, especially after fees (S&P Dow Jones Indices, n.d.). These findings reinforce the theoretical expectation that fees represent a direct cost to investors and therefore have a negative impact on net returns. Carhart (1997, p. 67-68) also shows that expense ratios, turnover and load fees are negatively related to abnormal mutual fund performance, which further supports the argument that costs are important when evaluating fund returns.

Some studies nuance this conclusion. Cremers and Petajisto (2009, p. 3330-3333) argue that certain "high active share" funds, those that significantly deviate from their benchmarks, may generate excess returns. This suggests that while fees are generally detrimental, their impact may depend on the degree and quality of active management.

Overall, the literature strongly supports a negative relationship between fees and investor returns, particularly when performance is measured net of costs. However, the extent to which fees can be justified by superior performance remains a debated issue.

2.4.2 Risk and Return

The relationship between risk and return is a cornerstone of financial theory. Markowitz (1952) introduced Modern Portfolio Theory, which posits that investors can achieve optimal portfolios by balancing risk and expected return. This framework implies a positive relationship between risk and expected returns, where higher risk should be compensated by higher potential rewards.

Empirical studies generally support the existence of a risk-return relationship, although the strength and consistency of this relationship vary across markets and time periods.(Sharpe, 1964, p. 425-426; Fama & French, 1992, p. 427-428) In the context of mutual funds, risk is often included as a control variable to ensure that differences in performance are not simply due to varying levels of risk exposure. (Jensen, 1968, p. 389-395; Carhart, 1997, p. 57-58) Nevertheless, some research suggests that higher risk does not always lead to higher realized returns, highlighting the importance of empirical testing in specific contexts. (Ang et al. 2006, p. 259-260)

In empirical fund studies, risk can be measured in several ways, including beta, standard deviation, downside risk or risk-adjusted performance measures such as the Sharpe ratio (Sharpe, 1966, p. 119-138; Simons, 1998, p. 33-37). This thesis does not treat the chosen risk-proxy as a substitute for these more precise statistical measures. Instead, the variable is used as a standardised proxy for the fund's general risk category, presented to Swedish retail investors.

2.4.3 ESG and Fund Performance

Sustainability considerations, commonly captured through ESG metrics, have become increasingly important in investment analysis. However, the relationship between ESG performance and financial returns remains somewhat ambiguous.

While Friede et al. (2015, p. 210-213) find that the majority of earlier studies report a non-negative relationship between ESG and corporate financial performance, more recent studies suggest that the financial effect depends on how sustainability is measured and implemented. Atz et al. (2023, p. 802-825) find that ESG integration appears to perform better than exclusion-based screening or divestment, but also show that the aggregate performance of ESG investing is often difficult to distinguish from conventional investing. This indicates that stronger ESG performance may be associated with better governance, lower regulatory risk and improved stakeholder relationships, but that these benefits do not automatically translate into higher realized returns for investors.

Similarly, Dimson et al. (2015, p. 3225-3228) emphasise the role of active ownership, suggesting that engagement in ESG issues can lead to improved firm performance over time. These findings support the view that ESG considerations can create value rather than merely impose constraints. Also, Sustainability has also become an increasingly important part of fund selection (Hartzmark and Sussman, 2019).

In contrast, other studies present a more critical perspective. Hong and Kacperczyk (2009) find that so-called "sin stocks" excluded by ethical investment criteria tend to outperform the market. This suggests that restricting the investment universe based on ESG criteria may lead to higher expected returns for non-ESG investments, potentially due to neglected risk premiums.

Investor preferences may also affect the expected returns of sustainable and less sustainable assets. Pástor et al. (2021, p. 550-552) argue that investor demand for green assets can increase their prices and reduce expected future returns, while less sustainable assets may offer higher expected returns as compensation for lower investor demand.

Taken together, the literature indicates that the relationship between ESG and fund performance is ambiguous. This makes ESG an important variable for further empirical investigation.

2.4.4 Index Funds versus Actively Managed Funds

The distinction between index funds and actively managed funds is central to the fund performance literature. Index funds aim to replicate the performance of a market index and are characterized by low fees and high diversification. In contrast, actively managed funds seek to outperform the market through stock selection and market timing, typically resulting in higher costs like higher fees (Malkiel, 1995, p. 550-552; Cremers and Petajisto, 2009, p. 3330-3331).

A large body of research finds that active fund performance is largely explained by common risk factors and costs rather than managerial skill (Carhart, 1997). Carhart's findings suggest that persistence in fund performance is limited and that high-performing funds do not consistently maintain their advantage over time.

This does not imply that all active funds are equally unable to create value. Cremers and Petajisto (2009) provide evidence that certain actively managed funds may outperform the index, particularly those with high active share. This indicates that while passive investing generally offers superior risk-adjusted returns, there may be exceptions within the universe of active funds.

Taken together, previous research generally suggests that index funds have a cost advantage, while active funds must generate sufficient gross outperformance to compensate for higher fees. This makes the distinction between index funds and actively managed funds central to this thesis, particularly when analysing both gross returns and net returns. See Appendix 6 for a summary of the previous research.

2.5 Positioning the Study and Justifying the Variables

The existing body of literature provides a strong foundation for understanding the determinants of fund performance, particularly with regard to fees, risk, and sustainability factors. While prior studies offer valuable insights, they also reveal important limitations that this thesis seeks to address.

Several influential studies, including Fama and French (2010), and Malkiel (1995), have discussed that actively managed funds tend to underperform passive index funds, particularly after accounting for fees. The findings strongly support the inclusion of fees as a key independent variable and highlight the importance difference between gross and net returns (Carhart, 1997). This thesis builds directly on this insight by incorporating both gross return and net return as dependent variables, allowing for a more nuanced analysis of performance from both a theoretical and investor-oriented perspective.

The inclusion of risk as a control variable is grounded in established financial theory (Markowitz, 1952; Sharpe, 1964). While these frameworks predict a positive relationship between risk and expected return, empirical findings in the mutual fund context have shown that this relationship is not always consistent (Fama & French, 1992; Ang et al. 2006). By controlling for risk level, this thesis ensures that observed differences in returns are not merely driven by varying levels of risk exposure, thereby improving the robustness of the analysis.

The role of sustainability, measured through ESG scores, represents a continuous research area. Studies such as Friede et al. (2015) suggest a generally positive or neutral relationship between ESG and financial performance, while other contributions, including Hong and Kacperczyk (2009), indicate that excluding certain high-return "sin stocks" may lead to lower returns. This difference in findings shows the relevance of including ESG as an independent variable without imposing a predetermined directional hypothesis. The thesis contributes to this debate by empirically testing the ESG-Return relationship within the specific context of Swedish funds.

While some studies have analysed these variables in combination, to the author's knowledge, relatively limited research has examined their interaction within the context of the Swedish fund market and across both index funds and actively managed funds. Prior research often examines these factors in separate strands of literature. For instance, studies on cost efficiency, such as Fama and French (2010), primarily focus on fees, while portfolio theory, including Markowitz (1952) and Sharpe (1964), emphasises risk. In parallel, ESG-related research, such as Friede et al. (2015), analyses sustainability as an independent factor. This represents a key limitation, as it does not fully capture the potential interactions between them. For example, the effect of fees may differ depending on the level of active management, while the impact of ESG may vary between index funds and actively managed funds. By integrating these variables within a single empirical framework, this thesis extends the existing literature and provides a more comprehensive analysis of the determinants of fund performance.

Another important contribution of this study lies in its geographical focus. Several influential studies on mutual fund performance are based on U.S. data, including Jensen (1968), Malkiel (1995), Carhart (1997), and Fama and French (2010). Although European fund markets have also been examined, for example by Otten and Bams

(2002), the Swedish fund market remains a relevant setting because investor behaviour, fee structures, regulation and fund composition may differ across countries. Data and analyses from Morningstar suggest that structural differences across markets, such as regulation, investor behavior, and fund composition, may influence performance outcomes (Morningstar, 2026a). By focusing on Sweden, this thesis adds to the limited body of research in the context of country-specific fund literature.

Finally, this thesis contributes by explicitly comparing index funds and actively managed funds within the same analytical framework. While previous studies have often compared index funds and actively managed funds, fewer have examined this comparison together with fees, risk levels and ESG scores within the same empirical framework. This study therefore contributes by analysing whether these fund characteristics are associated with fund performance, while also examining whether the relationship between ESG scores and returns differs between actively managed funds and index funds

This thesis builds on established theoretical and empirical research while addressing several identified gaps. While previous research, in the author's knowledge, has mostly compared actively managed funds and index funds, and separately examined the relationship between ESG characteristics and fund performance, fewer studies appear to have analysed whether the relationship between ESG performance and returns differs depending on fund type. Limited attention seems to have been given to whether sustainability characteristics influence index funds and actively managed funds differently.

Furthermore, previous fund performance studies often differ in whether they examine gross or net returns, with evidence showing that active fund performance appears stronger before expenses are deducted (Fama & French, 2010). Other studies rely on broad international samples, which provide important cross-country evidence but may be less focused on the specific characteristics of individual national markets such as Sweden (Cremers et al., 2016).

This study instead examines both gross and net returns within the Swedish fund market, allowing for a more investor-oriented analysis where management fees are directly incorporated into the performance measure.

By combining fund type, fee structure, ESG and risk proxy within a panel-data framework, this study aims to contribute to the existing literature through a more integrated comparison of the determinants of fund performance in the Swedish context.

2.6 Statistical Hypotheses

2.6.1 Hypothesis motivation

The hypotheses are based on the theoretical framework and previous empirical research. Prior studies suggest that actively managed funds often struggle to outperform passive index funds after fees, which motivates testing whether returns differ between index funds and actively managed funds (Malkiel, 1995; Fama & French, 2010). Risk is included because financial theory suggests a relationship between risk and expected return, while mutual fund studies commonly account for risk exposure when evaluating performance (Markowitz, 1952; Sharpe, 1964; Jensen, 1968; Carhart, 1997).

Sustainability ratings are included because previous ESG research shows mixed evidence regarding the relationship between sustainability and financial performance (Friede et al., 2015; Hong & Kacperczyk, 2009; Atz et al., 2023). Finally, the interaction between sustainability ratings and fund type is tested because ESG characteristics may affect index funds and actively managed funds differently, as active funds generally have greater flexibility in portfolio selection while index funds are more closely linked to benchmark composition.

2.6.2 Main Hypothesis

H0: There is no difference in fund returns between index funds and actively managed funds over the five-year period.

H1: There is a difference in fund returns between index funds and actively managed funds over the five-year period.

2.6.3 Sub-Hypotheses

Fees and Gross Fund Return

H0a: Fees have no relationship with gross fund returns over the five-year period.

H1a: Fees have a relationship with gross fund returns over the five-year period.

Sustainability Ratings and Fund Return

H0b: Sustainability ratings have no association with fund returns over the five-year period.

H1b: Sustainability ratings have an association with fund returns over the five-year period.

Sustainability Ratings and Fund Type

H0c: The association between sustainability ratings and fund returns does not differ between index funds and actively managed funds.

H1c: The association between sustainability ratings and fund returns differs between index funds and actively managed funds.

3. Scientific Methodology

This chapter presents the scientific methodology used in the study. The chapter explains the research philosophy, research design and research approach that guide the analysis.

3.1 Research Philosophy

Before conducting the empirical analysis, it is important to clarify the research philosophy of the study. The study is based on the assumption that fund returns, fees, ESG scores and risk levels can be measured and analysed statistically. This makes a quantitative research approach suitable for answering the research question. These assumptions determine how researchers understand the world and what is considered valid knowledge (Saunders et al., 2019, p. 101-102)

This study adopts a structured philosophical stance by considering three central dimensions: ontology, epistemology and axiology. Together, these perspectives form the foundation for the study's quantitative and empirical approach to analysing how fees, sustainability grades and risk levels affect fund returns in Sweden.

3.1.1 Ontology

Ontology concerns assumptions about the nature of reality (Collis & Hussey, 2021. p. 41-42). Within research, a distinction is commonly made between an objectivist perspective, where reality is considered external and independent of the researcher, and a subjectivist perspective, where reality is viewed as socially constructed (Collis & Hussey, 2021, p. 41-42).

This study adopts an objectivist ontological position, assuming that financial relationships, such as those between fees, risk, sustainability and fund returns, exist independently of the researcher. Fund performance, ESG scores and risk measures are treated as observable and measurable phenomena that reflect an underlying objective reality. This fits the study because the analysis is based on numerical fund data and regressions. The aim is to identify patterns in the fund sample that may also be relevant for similar Swedish funds, thereby contributing to broader insights into the Swedish fund market.

3.1.2 Epistemology

Epistemology concerns the nature of knowledge and how it can be obtained. A central distinction is made between positivism, which emphasises objective measurement and empirical testing, and interpretivism, which focuses on subjective meaning and social context (Bell et al., 2019, p. 15).

This study adopts a positivist epistemological perspective, which is aligned with its quantitative research design. From this perspective, knowledge is generated through the observation and analysis of measurable data, with the objective of identifying patterns, relationships and statistical relationships that can be tested statistically.

The use of numerical data on fund returns, fees, ESG scores and risk levels reflects this approach. By applying statistical methods, the study aims to produce findings that are objective, replicable and generalisable. While complete objectivity is difficult to

achieve, since choices such as variable selection and model specification are made by the authors, subjectivity is minimised through the use of standardised data sources and established analytical techniques.

3.1.3 Axiology

Axiology concerns the role of values in the research process. Within positivist research, the ideal is that the researcher maintains a value-neutral position and does not influence the results through personal beliefs or biases. (Collis & Hussey, 2021, p. 43; Saunders et al., 2019, p. 134-135).

This study adopts an objective axiological stance, aiming to separate the researchers personal views from the empirical analysis. Relationships between variables, such as the effect of sustainability grades on fund returns, are examined based on data rather than judgements.

At the same time, it is acknowledged that complete neutrality cannot be fully guaranteed, as methodological choices, including variable selection and model specification, may reflect underlying assumptions. However, through transparency and systematic application of methods, the study seeks to reduce potential bias and strengthen the credibility of its findings.

3.2 Research Design and Methodological Choice

The methodological choices are also based on previous mutual fund performance research. Prior studies commonly analyse fund performance by comparing returns across funds and relating performance to costs, risk factors and fund characteristics. Carhart (1997), for example, studies mutual fund performance using factor-based models and shows that expenses and common risk factors are important when explaining fund returns. Fama and French (2010) also examine mutual fund performance by distinguishing between gross and net returns, which is relevant for evaluating whether active managers create value before and after costs.

This thesis follows a similar logic by including both net return and gross return as dependent variables. Net return is the main dependent variable because it reflects the return investors receive after costs. Gross return is included as an alternative measure because it captures performance before fees. This distinction is important in fund performance research, since an actively managed fund may generate stronger gross returns but still deliver weaker net returns if the additional performance is smaller than the fees. Wermers (2000) also supports this distinction by decomposing mutual fund performance into components such as stock-picking ability, style, transaction costs and expenses.

Fees are included because fund expenses are central in the active-versus-passive debate. Previous studies show that costs are an important reason why actively managed funds often underperform passive alternatives after fees (Carhart, 1997; Fama and French, 2010). Barber et al. (2005) also show that expenses are relevant for investor behaviour in mutual funds. Therefore, including fees is necessary both from a theoretical perspective and from an investor perspective.

The active-versus-passive distinction is included because it is central in mutual fund performance research. Fama and French (2010) find that active managers, on average, struggle to generate enough benchmark-adjusted return to cover their costs. Cremers and Petajisto (2009) further show that the degree of active management matters, since funds with higher active share may differ more meaningfully from their benchmarks. In this thesis, fund type is included as a dummy variable to test whether actively managed funds differ from index funds in terms of performance.

ESG score is included because sustainability ratings have become an increasingly important part of fund evaluation. Friede et al. (2015) show that the relationship between ESG and financial performance is often positive or neutral, but not completely one-sided. Hartzmark and Sussman (2019) are especially relevant because they study Morningstar sustainability ratings and show that investors respond to sustainability information in mutual funds. Bollen (2007) also shows that socially responsible fund attributes can matter to investors beyond pure financial performance. This motivates the inclusion of ESG as an independent variable, while also recognising that the expected direction of the ESG-return relationship is theoretically ambiguous.

Risk level is included as a control variable because fund returns may partly reflect differences in risk exposure. The general importance of risk is supported by Modern Portfolio Theory and CAPM, where expected return is linked to risk (Markowitz, 1952; Sharpe, 1964). In fund performance research, risk is often controlled for using measures such as beta, volatility or factor exposures (Carhart, 1997). This thesis uses Avanza's standardised 1-7 risk category as a practical risk proxy. This measure is not as precise as beta or volatility, but it reflects the risk classification that Swedish retail investors encounter when comparing funds. Standardised risk information is required in key information documents under the EU PRIIPs framework.(EIOPA, n.d) The PRIIPs framework requires key information documents to provide investors with comparable information on investment products, including risks and costs (EIOPA, n.d). This makes the risk category relevant for this study because the EU states that consumer-friendly information should be provided about key features of investment products, including potential gains, risks and costs, and aims to improve transparency and comparability across the EU. (EIOPA, n.d). It is therefore used as an investor-facing risk control rather than as a complete market-based risk measure.

Monthly time fixed effects are included to control for common market conditions that affect all funds during the same month. Fund returns may be influenced by broad market movements, macroeconomic shocks, interest-rate changes or sector-wide conditions. By including time fixed effects, the study reduces the potential problem that the associations are driven by market movements rather than fund-characteristics. This approach is consistent with panel-data methodology, where time fixed effects are commonly used to control for shocks that affect all units in the same period (Wooldridge, 2010; Stock and Watson, 2020).

3.3 Research Approach

This study adopts a deductive research approach, whereby hypotheses are developed based on existing theory and subsequently tested using empirical data (Saunders et al., 2019, p. 153). The theoretical framework, drawing on concepts such as the Efficient Market Hypothesis, Modern Portfolio Theory and ESG theory, forms the basis for expectations regarding the relationships between variables.

Through statistical analysis, the study evaluates whether the empirical findings support or contradict these theoretical expectations. This approach makes it possible to assess the applicability of established financial theories within the context of the Swedish fund market.

The deductive approach is consistent with the study's positivist and objectivist stance, as it emphasises hypothesis testing, empirical observation and generalisation. By combining theory-driven expectations with quantitative analysis, the study contributes to both academic understanding and practical insights regarding fund performance.

3.4 Literature Search

The primary databases used were Umeå University Library and Google Scholar. In addition, DiVA was consulted at an early stage to identify previous theses, providing methodological guidance and contextual understanding.

The search process was guided by keywords related to the central themes of the study, including fund returns, fees, risk, sustainability grades, ESG, index funds and actively managed funds. To ensure both breadth and relevance, searches were conducted with and without filters. Within the Umeå University Library database, filters such as peer-reviewed publications, full-text availability and publication dates between 1990 and 2025 were applied. As Google Scholar does not provide a direct filter for peer-reviewed material, identified sources were manually verified through the university library.

In addition to keyword searches, a backward and forward snowballing approach was applied. This included reviewing reference lists of relevant articles, identifying frequently cited studies and exploring related research suggested by database algorithms. This process contributed to a systematic literature review, ensuring that the study is grounded in relevant and high-quality academic research.

Keywords: fund returns, fees, risk, sustainability grades, ESG, index funds, actively managed funds.

3.5 Source Criticism

Several measures were implemented to ensure a high standard of source criticism throughout the study. Particular emphasis was placed on the use of primary sources and peer-reviewed journal articles in order to strengthen the credibility. Relevance was also considered to reflect recent developments, while a limited number of older sources were included to support foundational theories.

The empirical data is collected from established financial data sources, including LSEG Workspace, Avanza and Morningstar. Gross return data is obtained from LSEG Workspace, while information on fees and fund characteristics is collected from Avanza. Sustainability scores are based on Morningstar data. Using these data sources makes the data more comparable across funds, since the variables are collected in a consistent way, although differences in provider methodology may affect variables such as ESG scores and risk classifications (Berg et al., 2022)

In a limited number of instances, websites were used as supplementary sources. Their use was restricted to well-established and reputable organisations, and all sources were critically evaluated to ensure their reliability and suitability for academic research.

3.6 Ethical and Societal Considerations

This thesis has been conducted in accordance with relevant ethical guidelines for academic work at Umeå University and with reference to the principles outlined in ALLEA (2023), *The European Code of Conduct for Research Integrity*. These guidelines emphasise the core principles of reliability, honesty, respect, and accountability as fundamental to ensuring high-quality, transparent, and unbiased research (ALLEA, 2023, p. 5).

AI was used during the thesis process as a support tool. The authors used AI to discuss ideas, receive feedback, and identify possible directions for literature search and earlier studies. AI was also used to suggest alternative formulations and to help make some arguments more precise.

The thesis work was written, revised and critically assessed by the authors. AI was not used to independently collect data, conduct the empirical analysis, produce regression results, make methodological decisions or draw conclusions. The use of AI was therefore limited to feedback and language support. The authors maintained academic independence throughout the process and remained fully responsible for the content and academic integrity of the thesis.

Since the study only uses publicly available or database-based fund data and does not involve human participants or personal data, certain ethical risks, such as issues related to informed consent or personal data protection, are not applicable. Nevertheless, the authors maintain responsibility for ensuring the integrity of the research process. This includes the careful selection of credible data sources, accurate data handling, and a consistent commitment to objectivity and transparency throughout the study.

The topic of sustainability and ESG considerations is inherently linked to normative and societal values. However, this thesis adopts a neutral and analytical perspective aligned with its research question. The purpose is not to argue that investors should choose one type of fund over another, but to examine how the selected variables are related to returns.

From a societal perspective, the findings of this study may have several implications. First, the results may contribute to how investors evaluate funds, particularly in relation to fees, risk exposure, and sustainability grades. By providing empirical evidence on how these factors relate to fund returns (both gross and net) the study may contribute to more informed investment decisions and improved capital allocation.

Second, fund management companies may be affected by the findings, especially in terms of how they structure fees, manage risks, and incorporate sustainability considerations into their investment strategies. If certain variables are shown to be significantly associated with returns, this may incentivize adjustments in fund design and management practices, particularly in the distinction between index funds and actively managed funds.

Finally, at a broader societal level, this research contributes to the understanding of how financial markets integrate sustainability, management, fee and risk considerations. By examining whether these variables are associated with financial performance in the Swedish context, the study adds to the wider discourse on the role of capital markets in supporting long-term economic objectives.

4. Practical Method

In this chapter the authors present how the empirical study is carried out in practice. It describes the sample, data collection, variables and regression models used in the analysis. The study is based on monthly observations for 84 Swedish funds over a five-year period. The chapter also explains how net return, gross return, fees, ESG scores, risk levels and fund type are used in the regressions.

4.1 Sample and Population

The sample consists of 84 Swedish funds with a focus on the Swedish market, of which 24 are index funds and 60 are actively managed funds. The sample includes both index funds and actively managed funds, which makes it possible to compare the two fund types. The sample includes all index funds and actively managed funds for which data on returns, fees, ESG scores and risk levels was available.

The analysis is based on monthly observations over a five-year period. The chosen period was April 2021 to April 2026. This allows the study to examine fund performance over time while maintaining consistency in the ESG measure used. The unit of analysis is the fund-month observation, meaning that each fund is observed once per month during the sample period.

Both index funds and actively managed funds are included because the comparison between these fund types is central to the research question. By using funds that operate within the same national market, the study improves comparability and reduces variation caused by differences in geographic exposure.

Overall, the selected sample provides a relevant basis for examining how fees, sustainability grades, and risk proxy are associated with fund returns in Sweden, while also enabling a systematic comparison between index funds and actively managed funds.

4.2 Regression Analysis

Regression analysis is used to examine whether fund returns are associated with fund type, fees, ESG scores and risk levels. The regressions are based on monthly observations for each fund in the sample. Since the study uses observational data, the results are interpreted as statistical relationships rather than causal effects (Stock & Watson, 2020, p. 49-50). Accordingly, this study does not aim to determine causality, but to analyse whether these variables are related to fund performance.

The empirical analysis is based on panel data, since the dataset contains repeated monthly observations for the same funds over time. A pooled Ordinary Least Squares model is first estimated as a baseline comparison. OLS estimates coefficients by minimising the sum of squared residuals and provides an initial benchmark for the relationship between the variables (Studenmund, 2014, p. 98). However, because the dataset follows several funds across multiple months, the main empirical analysis is conducted using random-effects panel regressions. This specification is appropriate because the study aims to analyse both variation across funds and variation over time, while retaining key between-fund variables such as fund type, fees and sustainability risk scores.

The random-effects model is used because several central independent variables vary mainly between funds rather than within the same fund over time. A fund fixed-effects model would absorb much of this cross-sectional variation and would therefore be less suitable for the main research question, which compares index funds and actively managed funds. The random-effects specification allows the analysis to account for the panel structure of the data while preserving the between-fund variation needed to estimate associations between fund characteristics and returns (Wooldridge, 2010, p. 169-176).

Monthly time fixed effects are included to account for time-specific shocks that may affect all funds simultaneously, such as macroeconomic conditions, market-wide movements and changes in investor sentiment. This reduces the risk that common monthly market conditions are incorrectly attributed to fund-specific characteristics (Stock and Watson, 2020, p. 361-381).

The panel structure also implies that observations within the same fund may be correlated over time. To address this issue, standard errors are clustered at the fund level. This adjustment improves inference by accounting for heteroskedasticity and within-fund correlation across monthly observations (Wooldridge, 2010, p. 49-76; Stock and Watson, 2020, p. 52-53).

While regression analysis relies on several assumptions, including linearity and exogeneity, these conditions may not fully hold in practice. In particular, unobserved fund characteristics and omitted variables may influence the estimated relationships. For this reason, the results are interpreted as associations rather than causal effects.

Several model specifications are used to assess whether the results are stable across alternative formulations. The baseline pooled OLS model provides an initial comparison. The main model estimates net returns using a random-effects panel specification. Additional models use gross return as an alternative dependent variable and include interaction terms to examine whether the relationship between sustainability risk, fees and returns differs between index funds and actively managed funds. If the results are similar across specifications, this strengthens the credibility of the findings. (Wooldridge, 2010, p. 169-179; Stock & Watson, 2020, p. 260-268).

4.3 Variables

This section presents the study's selection of dependent, independent, and control variables. Independent variables, also referred to as independent variables, are used to explain variation in the dependent variable, while control variables are included to account for additional factors that may influence the outcome but are not the primary focus of the analysis (Collis & Hussey, 2021, p. 77). Information regarding data sources is presented in the following chapter (Chapter 5).

4.3.1 Independent Variables

In line with the research question, the main independent variables in this study are fund type, fees and sustainability grades. The fund type is a dummy variable in the tests, which takes on the value of 1 if the fund is actively managed, and 0 if the fund is an index fund. These variables are selected based on their theoretical and empirical relevance in explaining fund performance, as previous research links fund returns to

costs, fund type and sustainability-related factors (Sharpe, 1964; Carhart, 1997; Fama & French, 2010; Friede et al., 2015).

Fees represent the cost of investing in a fund and are typically expressed as an annual percentage of assets under management. They are expected to negatively affect net returns, as higher costs reduce the overall performance received by investors (Carhart, 1997; Fama & French, 2010). As for gross return, the association is unknown. The fee data for the sample are collected from Avanza and refer to the total annual fee (Avanza, 2026). Furthermore, since returns are measured monthly, annual fees are converted into monthly fees before calculating net returns. This ensures that the fee adjustment is consistent with the monthly return.

$$Monthly\ fee = 1 - (1 - Yearly\ fee)^{1/12}$$

Sustainability grades are measured using Morningstar ESG scores. These scores reflect the sustainability risk of the fund's underlying holdings. ESG scores are widely used in both academic research and the financial industry as an indicator of sustainability risk and are increasingly incorporated into investment decision-making processes (Friede et al., 2015; Atz et al., 2023; UN Global Compact, 2025). The sustainability grades for the sample have been collected from Morningstar and are aggregated assessments of portfolio companies' environmental work, social responsibility and corporate governance. These scores are reported on a 0-100 scale, where lower values indicate lower sustainability risk. (Morningstar, 2026b).

4.3.2 Dependent Variables

The dependent variables in this study are gross fund return and net fund return. Fund return represents the overall financial performance of a fund over a given period, without adjusting for fees, and reflects the gross performance of the fund.

Net return, on the other hand, measures the return received by investors after deducting all relevant fees. This variable provides a more accurate representation of investor outcomes, as it directly incorporates the cost dimension associated with fund investments. By including both measures, the study is able to distinguish between theoretical performance and actual investor-relevant performance. The gross fund returns have been collected from LSEG Workspace on a monthly basis over the five year period. The net returns have been calculated using the following formula.

$$Monthly\ net\ return = (1 + Gross\ return) \times (1 - Monthly\ fee) - 1$$

The inclusion of both fund return and net return aligns with the study's objective of analysing how fees, sustainability grades, and risk levels affect performance, while also enabling a more nuanced comparison between index funds and actively managed funds.

4.3.3 Control Variable

To account for factors that may also be associated with fund returns, the study includes risk level as a control variable.

Including control variables helps improve the interpretation of the estimated relationships between fees, sustainability grades and fund returns (Wooldridge, 2010;

Stock & Watson, 2020). At the same time, the selection of variables is limited in order to maintain analytical clarity and focus on the main relationships of interest.

Risk level is used as a control variable to capture broad differences in fund risk profiles. The measure is based on Avanza's reported fund risk category, ranging from 1 to 7, where higher values indicate higher risk. This type of risk classification is designed to provide investors with a simplified and comparable indication of fund risk (Avanza, 2026; EIOPA, n.d.).

The use of this measure is justified by its consistency and availability across the sample, as well as its relevance in the Swedish retail fund market. The measure should not be interpreted as a precise statistical estimate of risk. Compared with beta, standard deviation or downside-risk measures, the risk category is less granular and may not fully capture all dimensions of financial risk. It is therefore used only as a control variable for broad risk proxies between funds.

4.3.4 Time horizon

A time horizon of five years has been selected for this study, using monthly data in order to obtain a large data set and enable more accurate regression results. Among the studies discussed in the problematization, one used a 15-year time horizon (Fortin & Michelson, 1999), one used ten years (Knutsson & Edlund, 2025), one five years (Piller & Harryzon, 2020) and one used three years (Fornander & Taipale, 2024). After careful consideration, the authors of this study decided to use a time horizon of five years like the study of Piller & Harryzon (2020), for two critical reasons. First, a long time perspective is preferred since funds have a long recommended investment horizon of 3-5 years for mixed funds and 5+ years of equity funds (Swedish Investment Fund Association, n.d.). Secondly, although sustainability grades were implemented by Morningstar in 2016, these started as a grade between 1-5 and were not standardized to today's sustainability grades until 2021 (Morningstar, 2021). Thus, a longer time perspective than five years would have rendered this variable inconsistent and risked distorted results.

4.4 Regression Models

To test the hypotheses, three regression models are estimated. The models examine how fund type, fees, sustainability ratings and risk level are associated with fund returns. Since the study uses monthly observations for the funds over time, a pooled panel specification with time fixed effects was selected since several independent variables, particularly fund type, fee risk levels and ESG scores, display limited to no variation over time within individual funds. A fixed-effects specification would therefore absorb much of the cross-sectional variation central to the research question. Time fixed effects were nevertheless included in order to control for market-wide shocks and macroeconomic conditions affecting all funds simultaneously.

The models use either net return or gross return as the dependent variable. Net return is used as the main dependent variable, since it reflects the return received by investors after fees. Gross return is included to see whether the results change when fees are not deducted.

In all models, i represents fund-ID and t represents month. $Active_i$ is a dummy variable equal to 1 for actively managed funds and 0 for index funds. Fee_i represents the total fund fee, ESG_i represents the Morningstar sustainability risk score, and $Risk_i$ represents the fund risk level. The γ_t represents monthly time fixed effects, while ϵ_{it} is the error term. Since the ESG score is measured as a sustainability risk score, a lower ESG value indicates a stronger sustainability profile.

4.3.1 Model 1: Random-effects panel model - Net return

$$NetReturn_{it} = \beta_0 + \beta_1 Active_i + \beta_2 Fee_i + \beta_3 ESG_i + \beta_4 Risk_i + \gamma_t + \epsilon_{it}$$

4.3.2 Model 2: Random-effects panel model - Gross return

$$GrossReturn_{it} = \beta_0 + \beta_1 Active_i + \beta_2 Fee_i + \beta_3 ESG_i + \beta_4 Risk_i + \gamma_t + \epsilon_{it}$$

4.3.3 Model 3: Random-effects panel model with ESG × Fund type interaction

$$NetReturn_{it} = \beta_0 + \beta_1 Active_i + \beta_2 Fee_i + \beta_3 ESG_i + \beta_4 (Active_i \times ESG_i) + \beta_5 Risk_i + \gamma_t + \epsilon_{it}$$

4.5 Truth Criteria - Reliability and Validity

Reliability concerns whether the study could produce similar results if it were repeated using the same data sources and method (Saunders et al., 2006, p. 149). In other words, a study is considered reliable if it produces stable and consistent findings over repeated applications.

In this study, the data is obtained from LSEG Workspace, Avanza and Morningstar. Gross return data is collected from LSEG Workspace, fee and fund characteristic data from Avanza, and sustainability scores from Morningstar. However, it should be noted that variables such as sustainability scores and risk measures may differ across databases, as they are constructed using different methodologies and underlying assumptions (Berg et al., 2022). Consequently, replicating the study using alternative data sources could lead to variations in results. Nonetheless, if the study were to be replicated using the same database and methodology, the reliability of the findings is considered to be high.

Validity concerns whether the study measures what it aims to measure. Two key dimensions of validity are typically distinguished: internal validity and external validity (Saunders et al., 2006, p. 150).

Internal validity relates to the ability of a study to establish causal relationships between variables and to determine the direction of such relationships (Saunders et al., 2006, p. 193). This study does not aim to establish causality between fees, sustainability grades,

risk levels, and fund returns. Instead, it focuses on identifying and analysing statistical relationships between these variables within the Swedish fund market. Since the study does not aim to establish causality, limitations related to causal inference are acknowledged but do not contradict the study's stated purpose of examining statistical associations (Stock & Watson, 2020, p. 49-50).

The validity of the risk variable requires particular consideration. Since the study uses a standardised retail risk category rather than directly estimated volatility or beta measures, the variable captures broad risk classification rather than exact financial risk exposure (EIOPA n.d.) This reduces measurement precision but does not invalidate the variable, as its purpose is to control for general differences in fund risk profiles across the sample.

External validity refers to the extent to which the findings of a study can be generalised beyond the specific context in which the research was conducted (Saunders et al., 2006, p. 151). This study is based on Swedish investment funds, with a particular focus on comparing index funds and actively managed funds. Given this context, the findings are primarily applicable to the Swedish fund market, which is characterized by specific institutional conditions, regulatory frameworks, and investor behaviors.

Caution should also be exercised when attempting to generalise the findings to other countries or financial markets with different structural characteristics. At the same time, the study adds evidence into how fees, sustainability grades, and standardised risk categories are associated with fund performance within a developed and sustainability-oriented financial market.

5. Data

This chapter presents the preparation and evaluation of the dataset prior to the empirical analysis. The chapter includes the screening process, construction of the final sample, correlation analysis and econometric diagnostics conducted before estimating the final regression models. Since the study is based on panel data with repeated monthly observations for each fund, several statistical tests are performed to evaluate whether the chosen model specifications are appropriate given the structure of the data.

5.1 Screening Process and Data Preparation

The data used in this study was collected from three primary sources. Gross monthly fund returns were extracted from LSEG Workspace. Further, both ESG scores and standardised risk proxy data were collected from Avanza, where sustainability scores are provided by Morningstar. Also, the yearly fund fee was collected from Avanza.

The initial sample consisted of Swedish mutual funds for which data on monthly returns, fees, ESG scores and risk levels were available. Both index funds and actively managed funds were included in the sample in order to enable comparisons between the two management strategies. The list of funds used was collected from Avanza by sorting on Swedish funds index funds and Swedish actively managed funds respectively.

Several screening steps were performed before constructing the final dataset. Funds with incomplete observations for one or more key variables were excluded from the sample. Although some funds belong to the same underlying management structure, they were retained since they represent separate investable products with different investor characteristics. On the other hand, this may increase similarity between certain observations and is therefore acknowledged as a limitation of the study. The initial sample consisted of 117 Swedish funds, a large number of these (33 funds) were eliminated because the data on their monthly returns were not available at LSEG Workspace for the corresponding time period.

The study applies monthly observations over a five-year period, resulting in a panel dataset where each fund is observed repeatedly over time. The unit of analysis is therefore the fund-month observation. The final dataset consists of 84 funds and approximately 5041 monthly observations.

Monthly net returns were calculated by adjusting gross monthly returns for the monthly equivalent of annual fund fees. Since annual fees are reported on a yearly basis, they were converted into monthly fees using the following formula:

The purpose of using net return as the primary dependent variable is that it better reflects the actual return received by investors after accounting for management costs.

The ESG variable consists of Morningstar sustainability scores ranging between 0 and 100, where lower values indicate stronger sustainability performance. Risk levels were obtained directly from Avanza and range from 1 to 7. A higher value indicates higher risk according to the standardised fund risk classification. (Avanza, 2026; EIOPA n.d.).

The dataset was structured in long panel format before being imported into Stata. Each row in the dataset represents one observation for one fund in one specific month. This

structure allows the study to estimate pooled panel regressions while accounting for repeated observations over time.

5.2 Correlation Analysis and Multicollinearity

Before estimating the regression models, the relationships between the independent variables were examined through correlation analysis. Excessively high correlations between independent variables may indicate multicollinearity problems, which can distort coefficient estimates and weaken the reliability of statistical inference. (Studenmund, 2014, p. 272)

To evaluate this issue, a correlation matrix was constructed for the independent variables included in the regression models.

Table 1. Correlations Matrix

	active	fee_mo~y	esg	risk
active	1.0000			
fee_monthly	0.7783*	1.0000		
	0.0000			
esg	0.2509*	0.4494*	1.0000	
	0.0000	0.0000		
risk	0.2950*	0.2447*	0.2483*	1.0000
	0.0000	0.0000	0.0000	

The correlation matrix showed no indications of excessively high correlations between the independent variables. None of the correlation coefficients approached the commonly referenced threshold of 0,80 (Studenmund, 2014, p. 272), suggesting that severe multicollinearity is unlikely to be present in the dataset. The only relatively strong correlation identified was between fee level and fund type. This is expected, since actively managed funds generally charge higher fees than index funds.

Table 2. VIF-Table

```
. vif

    Variable |       VIF       1/VIF
-------------+----------------------
    fee_total |      3.11    0.321196
active_dummy |      2.76    0.362564
         esg |      1.35    0.743134
        risk |      1.14    0.877468
-------------+----------------------
    Mean VIF |      2.09
```

The VIF results indicated low levels of multicollinearity across all independent variables. VIF values above 5 may indicate problematic multicollinearity (Studenmund, 2014, p. 273-274). Since none of the VIF values exceeded the accepted threshold levels, multicollinearity is not considered severe enough to affect the interpretation of the regression estimates.

5.3 Econometric Diagnostics

Since the study uses panel data consisting of repeated observations of the same funds over time, several econometric diagnostics were conducted prior to estimating the final regression models. These tests were performed in order to evaluate whether important assumptions behind the regression models were reasonably fulfilled.

5.3.1 Heteroskedasticity

One important assumption in OLS regression is homoskedasticity, meaning that the variance of the error term is constant across observations (Stock & Watson, 2020, p. 188-192). This is relevant because financial return data may display time-varying volatility and fat-tailed distributions (Cont, 2001). As an initial diagnostic, residual-versus-fitted plots were examined to assess whether the residuals displayed systematic patterns or a cone-shaped distribution, which may indicate heteroskedasticity or model misspecification (Stock & Watson, 2020, p. 150). The residual plots did not indicate any clear pattern of this. Additional residual diagnostics, including a histogram and Q-Q plot, showed that the residuals were centred around zero and approximately normally distributed, although some deviations were visible in the tails. These diagnostics are presented in Appendix 1-3.

A Breusch-Pagan test was then conducted as a formal test of heteroskedasticity (Breusch & Pagan, 1979). The test produced a chi-square value of 0.07 and a p-value of 0.7962 (see Appendix 4). Since the p-value is above the 5 percent significance level, the null hypothesis of constant variance cannot be rejected. This indicates that there is no strong evidence of heteroskedasticity in the baseline model.

5.3.2 Serial Correlation and Panel Structure

Because the dataset contains repeated monthly observations for the same funds, there is a possibility that observations within each fund are correlated over time. This issue is

commonly referred to as serial correlation or autocorrelation (Studenmund, 2014, p. 101).

Because the dataset contains repeated monthly observations for the same funds, there is a possibility of within-panel correlation in the error terms. Clustered standard errors are appropriate in panel data settings where observations within entities may be correlated over time.(Stock and Watson, 2020, p. 376) Therefore, all regression models apply standard errors clustered at the fund level.

5.3.3 Outliers and Distribution of Returns

Financial return data frequently contain extreme observations due to periods of unusually high market volatility (Cont, 2001). During the descriptive analysis, several observations with monthly fund returns exceeding approximately ±20 percent were identified.

The dataset was manually re-examined to determine whether these observations represented data entry errors or genuine market outcomes. Since multiple funds displayed similar fluctuations during the same periods, the observations were considered valid and retained in the final dataset. Maintaining these observations in the tests may result in higher residual variance and reduce the precision of the estimated coefficients.

Outliers should not automatically be removed unless there are clear reasons to believe that the observations are incorrect. (Studenmund, 2014, p. 78), Removing valid extreme observations may distort the dataset and reduce the representativeness of the analysis, particularly in financial research where volatility is an inherent characteristic of the data.

5.4 Final Model Specification

After evaluating the dataset and conducting the econometric diagnostics, the final empirical strategy consists of one baseline pooled OLS model and three main random-effects panel models. The pooled OLS model is used as an initial benchmark, while the main interpretation is based on the random-effects panel specifications.

The random-effects specification was chosen because the dataset contains repeated monthly observations for the same funds, while several key independent variables, including fund type, fee level, sustainability risk score and standardised risk level, vary mainly between funds. A fund fixed-effects model would therefore absorb much of the cross-sectional variation that is central to the research question, particularly the comparison between index funds and actively managed funds (Wooldridge, 2010, p. 169-176; Stock and Watson, 2020, p. 367-372).

Standard errors are clustered at the fund level to account for heteroskedasticity and within-fund correlation across monthly observations (Stock and Watson, 2020, p. 376).

The model may be affected negatively by issues related to endogeneity caused by omitted variables. If omitted factors are correlated with included variables, there is a chance that the OLS estimates may be influenced by the omitted variables as well. Thus, estimated coefficients should not be interpreted as causal effects, but rather as associations.

6. Empirical Results

6.1 Sample and sample distribution

The final sample consists of 84 Swedish funds of which 24 are index funds and 60 are actively managed funds. The full list is presented in Appendix 5.

Table 3. Sample distribution

active_dummy	Freq.	Percent	Cum.
0	1,441	28.59	28.59
1	3,600	71.41	100.00
Total	5,041	100.00	

Table 3 shows the distribution of fund types, where zero represents index funds and one represents actively managed funds. The sample for this study consists of 71,41% actively managed funds and the remaining 28,59% of the sample is made up of index funds. This sample distribution was not intended or organized by the authors, there is simply a much larger amount of actively managed funds than there are index funds in the Swedish fund universe. The distribution may lead to a higher degree of statistically significant findings for actively managed funds because of its larger sample size. Since monthly data over five years has been used, index funds also provide a large amount of data which mitigates this effect.

Table 4. Descriptive statistics

Variable	Obs	Mean	Std. dev.	Min	Max
net_return	5,041	.0031732	.0554267	-.2311366	.2679032
gross_return	5,041	.0041433	.0554156	-.2297934	.2691872
fee_total	5,041	.0115579	.0068401	0	.038
esg	5,041	20.05373	1.406586	16.09	23.75
risk	5,041	4.178536	.3830013	4	5
active_dummy	5,041	.714144	.451866	0	1

6.1.2 Net return

Net return is the main dependent variable in this study. It is calculated by adjusting each fund's monthly gross return for the corresponding monthly fee. The mean monthly net return is approximately 0.32%, with a standard deviation of approximately 5.5%. The minimum monthly net return during the sample period was around -23%, while the maximum was around 26%, which is relatively high considering these are funds and not individual stocks. These extreme values were re-examined to determine whether they reflected data-entry errors or valid market observations. Since several funds displayed

monthly fluctuations of around 20% during different periods, the observations were retained in the dataset.

6.1.3 Gross return

Gross return is used as an alternative dependent variable. The mean monthly gross return is approximately 0.4%, with a standard deviation of approximately 5.5%. The minimum gross return is roughly -23%, while the maximum is approximately 27%.

6.1.4 Total fee

The total fee is one of the independent variables in this study. The mean for the total fees is around 1,1% and has a standard deviation of approximately 0,68%. The minimum fee is 0% and the maximum fee is roughly 3,8%.

6.1.5 ESG score

The ESG score is the second independent variable of this study. The scores are based on Morningstar's sustainability risk measure and range from 0 to 100, where lower values indicate lower sustainability risk. The mean is around 20 with a standard deviation of 1,4. The minimum ESG score in the sample is 16,09 and the maximum is 23,75, thus indicating a relatively low spread within the sample which may affect the statistical significance of this variable.

6.1.6 Risk level

The risk variable represents the standardised fund risk proxy. It should be interpreted as an ordinal risk classification from 1 to 7 rather than a continuous market-based risk estimate, where a lower number means lower risk and lower potential reward. The mean risk proxy is approximately 4,1 with a standard deviation of 0,38. The minimum risk proxy in the sample is 4 and the maximum is 5. Similar to the ESG score, the spread is quite low in the risk proxy within the sample, which may affect its statistical significance.

6.1.7 Active management vs Index

Fund type is a binary variable used to compare actively managed funds and index funds. A value of 1 refers to actively managed funds, while a value of 0 refers to index funds.

Before investigating the potential association between the chosen variables, a comparison of the net return was conducted as initial exploratory analysis to create an overview of the index funds and actively managed funds in the sample.

Table 5. Two-sample t-test, net return

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
0	1,441	.0054858	.0013732	.0521284	.002792	.0081795
1	3,600	.0022475	.0009446	.0566738	.0003955	.0040994
Combined	5,041	.0031732	.0007807	.0554267	.0016427	.0047036
diff		.0032383	.0017274		-.0001481	.0066247

diff = mean(0) - mean(1) t = 1.8747
H0: diff = 0 Degrees of freedom = 5039

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = 0.9696 Pr(|T| > |t|) = 0.0609 Pr(T > t) = 0.0304

The two-sample t-test compares the mean net return between the two groups and is shown in Table 5. The index funds have a mean net return of approximately 0,55% per month and a standard deviation of around 0,13% whereas the actively managed funds show a mean net return of around 0,22% per month with a standard deviation of 0,009%. This means that when accounting for fund fees, index funds produce roughly 0,32% higher returns to the investor than actively managed funds. The test produced rather high p-values which means that the null hypothesis cannot be rejected at the 5% significance level. The alternative hypothesis (diff > 0) does produce a p-value of 0,03 which is significant at the 5% significance level, meaning that there is statistical support for index funds outperforming actively managed funds, although the test is one-sided and should be interpreted cautiously.

Table 6. Two-sample t-test, gross return

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
0	1,441	.0057463	.0013731	.052125	.0030527	.0084399
1	3,600	.0035016	.0009446	.0566736	.0016497	.0053536
Combined	5,041	.0041433	.0007805	.0554156	.0026132	.0056734
diff		.0022447	.0017273		-.0011417	.005631

diff = mean(0) - mean(1) t = 1.2995
H0: diff = 0 Degrees of freedom = 5039

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = 0.9031 Pr(|T| > |t|) = 0.1938 Pr(T > t) = 0.0969

Table 6 shows the comparison of the gross return between index funds and actively managed funds. Whereas the gross return is not directly linked to any of the statistical hypotheses, the findings are quite interesting. The mean gross return for index funds is around 0,57%, which is very close to the mean net return of 0,55% shown in figure 3. The actively managed funds produce a mean gross return of 0,35%, which is 0,13 percentage points higher than their mean net return. The t-test results suggest that index funds have higher average returns than actively managed funds in the sample. However, the difference is not statistically significant at the 5% level in the two-sided test, meaning that the result should be interpreted as descriptive rather than conclusive evidence.

The t-test results suggest that index funds have higher average returns than actively managed funds in the sample. However, the difference is not statistically significant at the 5% level in the two-sided test, meaning that the result should be interpreted as descriptive rather than conclusive evidence.

6.2 Baseline pooled OLS model

The baseline model is a pooled OLS regression and is used as the first regression specification in the study. The model includes 5041 observations and examines the relationship between net return and active management, total fee, ESG score and risk level. This model provides an initial estimate of the associations between the independent variables and monthly net returns.

Table 7. Baseline pooled OLS model

Linear regression

Number of obs	= 5,041
F(4, 5036)	= 2.69
Prob > F	= 0.0295
R-squared	= 0.0023
Root MSE	= .05539

net_return	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
active_dummy	-.0035908	.0027671	-1.30	0.194	-.0090156	.001834
fee_total	.152734	.1937384	0.79	0.431	-.2270775	.5325456
esg	-.0017454	.0006605	-2.64	0.008	-.0030403	-.0004504
risk	-.0003549	.0024691	-0.14	0.886	-.0051953	.0044855
_cons	.0404561	.0145597	2.78	0.005	.0119128	.0689994

6.2.1 Model fit

The model is statistically significant at the 5% level, with a Prob > F value of 0,0295. The R-squared value is 0,0023, which means that the model explains a small part of the variation in monthly net returns. The low R-squared indicates that the included fund characteristics explain only a limited share of short-term monthly return variation. This

is not unexpected, since monthly fund returns are likely to be strongly influenced by market-wide movements not captured by the model.

6.2.2 Active management

The coefficient for active management is negative at -0,0036. This means that actively managed funds have lower net returns than index funds in the baseline model. The result is not statistically significant, as the p-value is 0,194.

6.2.3 Total fee

The coefficient for total fee is positive at 0,1527. This means that the model shows a positive relationship between total fee and net return. The relationship is not statistically significant, with a p-value of 0,431.

6.2.4 ESG score

The ESG coefficient is negative at -0,0017 and statistically significant at the 1% level. This means that the ESG score has a statistically significant relationship with net return in the baseline model. Since the ESG variable is a sustainability risk score, lower values indicate lower sustainability risk. The negative coefficient therefore implies that higher ESG risk scores are associated with lower net returns, or equivalently, that funds with lower sustainability risk are associated with higher net returns.

6.2.5 Risk level

The coefficient for the standardised risk categories is negative at -0,0004. This result is not statistically significant, as the p-value is 0,886. Therefore, the baseline model does not show a statistically significant relationship between the standardised risk proxy and net return.

6.2.6 Summary of baseline model

Overall, the baseline model shows that ESG score is the only statistically significant independent variable. Active management, total fee and standardised risk proxy are not statistically significant in this model.

6.3 Random-Effects Panel Model - Net return

The random-effects panel model applies the same specification as the baseline OLS model while additionally accounting for the panel structure of the dataset. Since the study consists of repeated monthly observations for the same funds over time, the random-effects specification allows the analysis to account for unobserved fund-specific effects while retaining both within- and between-fund variation.

The model includes 5041 observations across 84 funds and applies standard errors clustered at the fund level. Compared to the baseline pooled OLS model, the random-effects specification is the main panel-data approach used in this thesis. It accounts for the repeated observation structure of the dataset while retaining both within-fund and between-fund variation, which is important because several key

independent variables vary mainly between funds (Wooldridge, 2010, p. 169-176; Stock and Watson, 2020, p. 367-372).

Table 8. Random-Effects Panel Model - Net return

Random-effects GLS regression — Number of obs = 5,041
Group variable: id — Number of groups = 84

R-squared:
Within = 0.0000
Between = 0.4578
Overall = 0.0023

Obs per group:
min = 60
avg = 60.0
max = 61

Wald chi2(4) = 181.92
corr(u_i, X) = 0 (assumed) — Prob > chi2 = 0.0000

(Std. err. adjusted for 84 clusters in id)

net_return	Coefficient	Robust std. err.	z	P>|z|	[95% conf.	interval]
active_dummy	-.0035908	.0008677	-4.14	0.000	-.0052914	-.0018902
fee_total	.152734	.0669532	2.28	0.023	.0215081	.2839599
esg	-.0017454	.0002321	-7.52	0.000	-.0022003	-.0012904
risk	-.0003549	.0007416	-0.48	0.632	-.0018083	.0010985
_cons	.0404561	.0053554	7.55	0.000	.0299598	.0509525
sigma_u	0					
sigma_e	.05574874					
rho	0	(fraction of variance due to u_i)				

6.3.1 Model fit

The random-effects model is statistically significant at the 1 percent level, with a Prob > chi2 value of 0,0000. Similar to the baseline model, the overall R-squared value is relatively low at 0,0023, meaning that the independent variables explain only a small proportion of the variation in monthly net returns.

Low explanatory power is common in monthly return regressions, since fund performance is heavily influenced by short-term market movements and macroeconomic shocks outside the model specification. The between R-squared value of 0,4578 suggests that the independent variables explain a substantially larger share of the variation across different funds than within individual funds over time.

6.3.2 Active management

The coefficient for active management is negative at -0,0036 and statistically significant at the 1 percent level. This indicates that actively managed funds are associated with lower monthly net returns than index funds within the random-effects specification.

Compared to the baseline OLS model, where the coefficient was not statistically significant, the random-effects model strengthens the evidence that index funds outperform actively managed funds during the sample period when controlling for fees, ESG score and risk level.

6.3.3 Total fee

The coefficient for total fee is positive at 0,1527 and statistically significant at the 5 percent level. This indicates a positive relationship between fee level and monthly net returns within the sample. The positive coefficient is unexpected in relation to prior fee-performance literature and is therefore interpreted cautiously.

6.3.4 ESG score

The ESG coefficient is negative at -0,0017 and statistically significant at the 1 percent level. Since lower ESG scores indicate stronger sustainability performance, the negative coefficient implies that funds with higher sustainability risk scores are associated with lower monthly net returns.

The result therefore suggests that stronger sustainability performance is associated with higher investor returns during the sample period. Compared to the baseline OLS model, the ESG coefficient remains statistically significant, strengthening the robustness of the relationship.

6.3.5 Risk level

The coefficient for risk proxy is negative at -0,0004. The relationship is not statistically significant, as the p-value is 0,632.

6.3.6 Summary of the random-effects model

Overall, the random-effects model shows statistically significant relationships between active management, fee level, ESG score and monthly net returns..

Compared to the baseline OLS model, the random-effects specification strengthens several of the observed associations, particularly regarding active management and fee level. This indicates that accounting for the panel structure of the dataset influences the estimated significance of the independent variables.

6.4 Random-Effects Panel Model - Gross return

The gross return model applies the same random-effects specification as the previous model but uses gross return instead of net return as the dependent variable. The purpose of this model is to examine whether the relationships between active management, fee level, ESG score and risk level differ when fund performance is measured before fees are deducted. This allows for a comparison between gross return and investor-relevant net return.

Similar to the previous specification, the model accounts for the panel structure of the dataset through a random-effects framework and applies standard errors clustered at the fund level. The model includes 5041 observations across 84 funds.

Table 9. Random-Effects Panel Model - Gross return

Random-effects GLS regression — Number of obs = 5,041
Group variable: id — Number of groups = 84

R-squared:
Within = 0.0000
Between = 0.4107
Overall = 0.0019

Obs per group:
min = 60
avg = 60.0
max = 61

Wald chi2(4) = 132.13
corr(u_i, X) = 0 (assumed) — Prob > chi2 = 0.0000

(Std. err. adjusted for 84 clusters in id)

gross_return	Coefficient	Robust std. err.	z	P>|z|	[95% conf.	interval]
active_dummy	-.0035978	.0008679	-4.15	0.000	-.0052989	-.0018966
fee_total	.2374537	.0669907	3.54	0.000	.1061544	.368753
esg	-.0017456	.0002321	-7.52	0.000	-.0022006	-.0012906
risk	-.0003551	.0007417	-0.48	0.632	-.0018089	.0010987
_cons	.0404576	.0053561	7.55	0.000	.0299597	.0509554
sigma_u	0					
sigma_e	.05574874					
rho	0	(fraction of variance due to u_i)				

6.4.1 Model fit

The random-effects gross return model is statistically significant at the 1 percent level, with a Prob > chi2 value of 0,0000. The overall R-squared value is 0,0019, indicating that the independent variables explain a relatively small proportion of the variation in monthly gross returns, as expected in a financial return regression

The between R-squared value of 0,4107 indicates that the independent variables explain a substantially larger share of the variation across funds than within individual funds over time.

6.4.2 Active management

The coefficient for active management is negative at -0,0036 and statistically significant at the 1 percent level. This indicates that actively managed funds are associated with monthly gross returns than index funds within the random-effects specification.

The result suggests that the lower returns associated with actively managed funds are also present when performance is measured before fees are deducted. Consequently, the lower performance of actively managed funds in the sample does not appear to be driven solely by higher fee levels.

6.4.3 Total fee

The coefficient for total fee is positive at 0,2375 and statistically significant at the 1 percent level. This indicates a positive relationship between fee level and monthly gross returns.

Compared to the net return model, the coefficient for fee level is larger in the gross return specification. This is expected since gross returns measure performance before fees are deducted.

6.4.4 ESG score

The ESG coefficient is negative at -0,0017 and statistically significant at the 1 percent level. Since lower ESG scores indicate stronger sustainability performance, the result suggests that funds with stronger sustainability profiles generate higher monthly gross returns during the sample period.

The coefficient remains highly similar in both magnitude and significance compared to the net return specification. This strengthens the indication that the relationship between sustainability performance and fund returns is relatively robust regardless of whether returns are measured before or after fees.

6.4.5 Risk level

The coefficient for the risk proxy is negative at -0,0004 and statistically insignificant, with a p-value of 0,632. Consequently, the model does not provide evidence of a statistically significant relationship between the standardised risk proxy classification and monthly gross fund returns.

Similar to the net return model, this may indicate that the risk classification does not fully capture the type of short-term volatility associated with monthly return fluctuations, or that the variation in risk levels across the sampled funds is relatively limited.

6.4.6 Summary of the random-effects model - Gross return

Overall, the random-effects gross return model produces results that are highly similar to the net return specification. Active management and ESG score remain negatively associated with fund returns, while fee level shows a positive and statistically significant relationship with gross performance.

The similarities between the gross and net return models suggest that the observed relationships are not solely driven by the mechanical effect of deducting fees from returns. This indicates that index funds show higher gross returns than actively managed funds in the model.

6.5 Random-effects panel model with ESG × Fund Type Interaction

The interaction model extends the previous random-effects specification by including an interaction term between active management and ESG score. The purpose of the model is to examine whether the relationship between ESG score and monthly net returns differs between actively managed funds and index funds. This directly addresses the comparative objective of the study by testing whether sustainability characteristics influence the two fund types differently.

The model includes 5041 observations across 84 funds and applies clustered standard errors at the fund level in order to account for the panel structure of the dataset.

Table 10. Random-Effects - Interaction Model

```
. xtreg net_return active_dummy esg interaction fee_total risk, re vce(cluster
```

Random-effects GLS regression		Number of obs	= 5,041
Group variable: id		Number of groups	= 84
R-squared:		Obs per group:	
Within = 0.0000		min	= 60
Between = 0.4685		avg	= 60.0
Overall = 0.0023		max	= 61
		Wald chi2(5)	= 342.91
corr(u_i, X) = 0 (assumed)		Prob > chi2	= 0.0000

(Std. err. adjusted for 84 clusters in id)

net_return	Coefficient	Robust std. err.	z	P>|z|	[95% conf.	interval]
active_dummy	-.0184803	.0061149	-3.02	0.003	-.0304654	-.0064952
esg	-.0023408	.0001982	-11.81	0.000	-.0027293	-.0019523
interaction	.0007672	.0003164	2.42	0.015	.000147	.0013875
fee_total	.1380425	.0680432	2.03	0.042	.0046802	.2714048
risk	-.0004709	.0007257	-0.65	0.516	-.0018933	.0009515
_cons	.0525746	.0047703	11.02	0.000	.0432251	.0619242
sigma_u	0					
sigma_e	.05574874					
rho	0	(fraction of variance due to u_i)				

6.5.1 Model fit

The interaction model is statistically significant at the 1 percent level, with a Prob > chi2 value of 0,0000. Similar to the previous specifications, the overall R-squared value remains relatively low at 0,0023. As discussed previously, this is not unexpected and the independent variables account for only a limited proportion of the variation in monthly returns.

The between R-squared value is substantially higher at 0,4685, suggesting that the independent variables are more effective at explaining differences across funds than short-term monthly fluctuations within individual funds over time. This is reasonable given that variables such as fee level, ESG score and fund type are relatively stable characteristics that vary more between funds than within funds over shorter time horizons.

6.5.2 Active management

The coefficient for active management is negative at -0,0185 and statistically significant at the 1 percent level. Because the model includes an interaction between active management and ESG score, the coefficient for active management should be interpreted as the difference between actively managed funds and index funds when the ESG score equals zero.

Since this value lies outside the observed range of ESG scores, the standalone active coefficient is not substantively meaningful on its own. The interaction term and the combined marginal effect are therefore more relevant for interpretation.

6.5.3 Total fee

The coefficient for total fee is positive at 0,1380 and statistically significant at the 5 percent level. This indicates that higher fee levels are associated with higher monthly net returns within the sample.

The positive relationship between fees and returns contradicts much of the previous literature, which often argues that higher fees reduce investor returns over time. One possible explanation is that higher-fee funds pursue investment strategies associated with higher gross performance potential or greater market exposure. Another possible explanation is that fee level partly captures differences in fund management style not fully controlled for in the model.

The relationship should therefore be interpreted cautiously, particularly since the result may also reflect characteristics specific to the sampled funds rather than a general relationship applicable across the broader fund market.

6.5.4 ESG score

The ESG coefficient is negative at -0,0023 and statistically significant at the 1 percent level. Since lower ESG scores represent stronger sustainability performance, the negative coefficient indicates that funds with stronger sustainability profiles are associated with higher monthly net returns.

Within this specification, the ESG coefficient represents the relationship between ESG score and net return for index funds, since index funds constitute the reference category in the interaction model.

The result therefore suggests that ESG performance has a statistically significant positive relationship with returns among index funds during the sample period.

6.5.5 Risk level

The coefficient for the risk proxy is negative but statistically insignificant, with a p-value of 0,516. Consequently, the model does not provide evidence of a statistically significant relationship between the Avanza risk classification and monthly net returns.

One possible explanation is the relatively limited variation in risk classifications within the sample, where most funds are concentrated within a narrow range of standardised risk proxies.

6.5.6 Interaction effect

The interaction term between active management and ESG score is positive at 0.0008 and statistically significant at the 5 percent level. Since the ESG variable is a sustainability risk score, the negative ESG coefficient for index funds indicates that

lower sustainability risk is associated with higher net returns. The positive interaction term shows that this ESG-return relationship is weaker for actively managed funds.

The ESG slope is -0.0023 for index funds and approximately -0.0015 for actively managed funds when the interaction effect is added. This suggests that lower sustainability risk is more strongly associated with returns among index funds than among actively managed funds in the sample.

6.5.7 Summary of the Interaction model

Overall, the interaction model strengthens the evidence that index funds outperform actively managed funds during the sample period. The results further suggest that stronger sustainability performance is associated with higher net returns, particularly among index funds.

The statistically significant interaction effect indicates that the relationship between ESG score and fund performance differs between actively managed funds and index funds. This supports the study's comparative research objective and suggests that the association differs across fund types.

6.6 Additional Robustness Analysis: Fee/Fund Type Interaction

The previous regression models identified an unexpected positive relationship between fee level and monthly net returns. This finding contradicts much of the previous literature within fund performance research, where higher fees are generally expected to reduce investor returns after costs (Carhart, 1997; Fama & French, 2010). Since actively managed funds typically charge substantially higher fees than index funds, there was a possibility that the positive fee coefficient reflected differences between fund types rather than a direct relationship between fees and returns.

To further examine this issue, an additional robustness regression was estimated including an interaction term between fee level and fund type. The purpose of this specification was to investigate whether the relationship between fee level and monthly net return differs between actively managed funds and index funds.

Table 11. Robustness Regression - Fee Interaction Effect

Random-effects GLS regression — Number of obs = 5,041
Group variable: id — Number of groups = 84

R-squared:
Within = 0.0000
Between = 0.4788
Overall = 0.0024

Obs per group:
min = 60
avg = 60.0
max = 61

Wald chi2(6) = 416.56
corr(u_i, X) = 0 (assumed) — Prob > chi2 = 0.0000

(Std. err. adjusted for 84 clusters in id)

net_return	Coefficient	Robust std. err.	z	P>|z|	[95% conf.	interval]
active_dummy	-.0138862	.0061653	-2.25	0.024	-.0259699	-.0018024
esg	-.0020266	.0002026	-10.00	0.000	-.0024236	-.0016296
fee_total	-.4486836	.2359113	-1.90	0.057	-.9110613	.0136941
interaction	.0004245	.0003357	1.26	0.206	-.0002334	.0010825
risk	-.0004564	.00073	-0.63	0.532	-.0018872	.0009743
active_dummy#c.fee_total						
1	.6051192	.2459321	2.46	0.014	.1231012	1.087137
_cons	.048222	.0045703	10.55	0.000	.0392644	.0571795
sigma_u	0					
sigma_e	.05574874					
rho	0	(fraction of variance due to u_i)				

The robustness regression shows that the direct coefficient for fee level becomes negative at -0,4487 and marginally insignificant at the 5 percent level, with a p-value of 0,057. This indicates that higher fees are associated with lower net returns among index funds, which is more consistent with traditional financial theory and previous empirical research as discussed in the literature review.

At the same time, the interaction term between fee level and active management is positive at 0,6051 and statistically significant at the 5 percent level. This suggests that the relationship between fees and returns differs significantly between actively managed funds and index funds. More specifically, the positive relationship between fee level and net return appears to be primarily driven by actively managed funds.

The result should be interpreted cautiously, since fee level is closely related to fund type and may capture differences in investment strategy, management style or market exposure that are not fully controlled for in the model. Overall, the robustness analysis suggests that the relationship between fees and fund performance is more nuanced than indicated in the previous models.

6.7 Evaluation of the Hypotheses

Table 12. Evaluation of the Hypotheses

Hypothesis	Result	Interpretation
There is a statistically significant difference in fund returns between index funds and actively managed funds.	Supported	The results indicate statistically significant differences between fund types, with index funds associated with higher returns in the sample. .
Sustainability ratings have a statistically significant relationship with fund returns.	Supported	Stronger ESG performance, meaning stronger sustainability profiles, are associated with higher returns.
The relationship between sustainability ratings and fund returns differs between index funds and actively managed funds.	Supported	The interaction term indicates that the ESG-return association differs by fund type.
Fees are associated with fund returns.	Nuanced	The association is statistically significant in several models, but the positive sign and interaction results suggest that the relationship depends on fund type and may reflect broader fund characteristics.

The hypotheses can be evaluated based on whether the empirical results support or do not support the expected associations. The strongest support is found for the active-passive comparison and the ESG association. Fees are also relevant, but the interpretation is more complicated because fee levels may reflect broader fund characteristics. The weakest support is found for standardised risk level, which does not explain monthly returns in the models.

7. Analysis

The purpose of this chapter is to interpret and discuss the empirical findings in relation to the theoretical framework and previous literature presented earlier in the thesis. The chapter analyses whether the observed relationships support the developed hypotheses and discusses potential economic explanations for the findings. Since the study is based on observational panel data, the findings are interpreted as statistical associations rather than causal relationships.

7.1 Active Management vs Index Funds

The results indicate a difference between actively managed funds and index funds, particularly in the random-effects specifications. In the main model, actively managed funds are associated with lower returns than index funds. This is one of the most important findings of the study, since the active-passive comparison is central to the research question.

From a theoretical perspective, this result is in line with the Efficient Market Hypothesis (Fama, 1970). If markets are relatively efficient, it should be difficult for active fund managers to consistently identify mispriced assets or time the market successfully. Active management can still add value in theory, but it needs to generate enough additional return to compensate for higher fees and manager-specific risk. In this study, the results do not indicate that actively managed funds generated sufficient additional returns to compensate for higher costs on average.

The finding also fits well with the cost-efficiency argument. Index funds are usually built to follow a benchmark at a low cost, while actively managed funds charge higher fees because they offer stock selection, market analysis and manager judgement (Fama and French, 2010). Consequently, the higher costs need to be matched by higher performance. In the results, the active funds do not appear to deliver enough additional performance to justify this difference. This supports previous research by Malkiel (1995), Carhart (1997), and Fama and French (2010), where active funds often struggle to outperform index funds after costs.

At the same time, the result should not be interpreted as proof that all active management is unsuccessful. Some actively managed funds may perform well during certain market periods or in specific market segments. The result is instead an average relationship in this sample. It shows that, when the funds are compared as groups, index funds performed better during the analysed period.

The finding may also be connected to the idea of closet indexing. Cremers and Petajisto (2009) argue that some active funds stay close to their benchmark while still charging active-management fees. If some of the active funds in this sample have relatively benchmark-like portfolios, this could help explain why they do not outperform index funds. They may carry higher costs without creating enough differentiated exposure.

Overall, the findings support the main hypothesis that fund returns differ between index funds and actively managed funds. The evidence points in favour of index funds in the

Swedish sample, although the result should be interpreted as an average association rather than evidence that all active management is unsuccessful.

7.2 ESG and Fund Performance

The coefficient for ESG score is negative and statistically significant in the main models. Since the Morningstar sustainability risk score works in the opposite direction of a traditional rating, this means that lower sustainability risk scores are associated with higher fund returns

One argument in theory is that stronger ESG performance can be connected to better risk management, stronger governance and lower exposure to regulatory or reputational problems (Friede, et al., 2015). From this perspective, ESG may capture something financially relevant about the companies. This interpretation is consistent with Friede et al. (2015), who found that many previous studies show a positive or neutral relationship between ESG and financial performance.

On the other hand, ESG restrictions may reduce the investment universe and exclude firms or sectors that could generate high returns. Hong and Kacperczyk (2009), for example, show that controversial "sin stocks" may offer higher expected returns because some investors avoid them for social or ethical reasons. This suggests that sustainability screening could theoretically create a return trade-off. The results in this thesis do not support that concern in this sample. Instead, stronger sustainability profiles are associated with higher returns during the analysed period.

Still, the result should be interpreted carefully. The study does not prove that ESG causes higher returns. It shows that ESG scores are statistically associated with fund returns after controlling for fund type, fees and standardised risk proxy. This matters because ESG scores may also reflect other fund characteristics. For example, funds with stronger sustainability profiles may have different sector exposure, geographic exposure or investment style compared to funds with weaker sustainability profiles. These factors may also explain the association.

The result is still relevant for the thesis because it shows that ESG is not only a descriptive label. In this sample, ESG has explanatory value in the regression models. This also supports the decision to include sustainability as a central variable.

In relation to the hypothesis on sustainability ratings and fund returns, the result supports the alternative hypothesis. Sustainability ratings are statistically associated with fund returns. Since lower ESG scores indicate stronger sustainability profiles, the results indicate an association between stronger sustainability performance and higher returns.

7.3 Differences Between Active and Passive Funds in ESG Performance

The interaction term between active management and ESG score is positive and statistically significant in the interaction model. Since a lower ESG score means stronger sustainability performance, this means that the ESG-return association is

weaker among actively managed funds than among index funds. This is an interesting result because it shows that the ESG-return association is not the same across the two fund types.

One possible interpretation is that index funds capture sustainability-related exposure more efficiently. If sustainability-related information is already reflected in market prices, index funds with stronger ESG profiles may benefit from broad exposure to companies with lower sustainability risk. They do not need to rely on manager skill to create this exposure.

For actively managed funds, ESG is only one part of the investment strategy. These funds also depend on the manager's choices, such as which companies, sectors and markets to invest in. This may explain why the link between ESG score and return is weaker for active funds. A strong sustainability profile may help, but the fund's final return still depends on whether the manager makes good investment decisions.

This interpretation connects directly to the active management debate in the theoretical framework. Active management is justified by the assumption that fund managers can identify mispriced securities or construct portfolios that outperform the market. Carhart (1997) shows that persistence in mutual fund performance is largely explained by common risk factors and investment expenses, which raises doubts about the ability of active managers to consistently create superior returns. Similarly, Fama and French (2010) found limited evidence that active mutual funds generate enough performance to cover their costs. These findings are relevant for interpreting why the ESG-return relationship may be weaker among active funds: even if sustainability characteristics are favourable, they may be offset by fees, unsuccessful stock selection or other active management decisions.

This result may also be explained by closet indexing. Closet indexing means that an active fund is managed almost like its benchmark, but still charges active-management fees (Cremers and Petajisto, 2009). If some active funds in the sample are close to their benchmarks, they may not be different enough from index funds for ESG to matter strongly for performance. This means that investors may pay higher fees without getting a clearly different strategy. Therefore, active funds need to be truly active if they are going to justify their higher fees.

This interpretation may also be linked to research suggesting that ESG characteristics are related to firm-specific risk. For example, Xu et al. (2025) find that ESG performance is associated with idiosyncratic and extreme risk at the stock level. Since their study concerns individual stocks rather than mutual funds, the connection should be interpreted cautiously. In this thesis, the result may indicate that ESG characteristics are more closely related to resilience or risk exposure than to higher returns alone. This is especially interesting because of the turbulence in the financial markets following events like COVID 19, the Ukraine-conflict and heightened tariffs from the US, to name a few. The implication may be that the true difference lies in a stronger resilience from ESG focused performance, rather than higher overall returns. The interaction result should not be interpreted as evidence that ESG causes higher returns for index funds. It may also reflect differences in benchmark composition, sector exposure or fund selection within the Swedish market.

In relation to the hypothesis concerning sustainability ratings and fund type, the findings support the alternative hypothesis that the relationship between sustainability ratings and fund returns differs between index funds and actively managed funds. The statistically significant interaction term indicates that the ESG-return relationship is not the same across the two fund categories in the sample. The null hypothesis that the relationship does not differ between fund types is therefore not supported.

7.4 Fees and Fund Performance

The fee results are more difficult to interpret than the ESG and active-management results. From a theoretical perspective, fees should reduce net returns. This is one of the clearest arguments in favour of index funds: if two funds provide similar exposure, the lower-fee fund should leave more return for the investor. This is also consistent with Malkiel (1995) and Fama and French (2010), who show that costs are a major reason why actively managed funds often underperform.

This theoretical expectation is supported by Malkiel (1995), who finds that equity mutual funds tend to underperform after accounting for costs. Fama and French (2010) also show that the aggregate portfolio of actively managed mutual funds is close to the market portfolio, but that the higher costs of active management reduce investor returns. These studies are important because they suggest that active managers must generate enough additional gross return to compensate investors for the extra cost.

The Efficient Market Hypothesis strengthens this expectation. If markets are relatively efficient, active managers should struggle to consistently generate abnormal returns through stock selection or market timing (Fama, 1970). This theoretical logic is central to the comparison between actively managed funds and index funds, since active funds generally charge higher fees while index funds are designed to provide broad market exposure at lower cost.

The empirical results in this thesis are not as simple as “higher fees lead to lower returns”. In the main models, total fee is positively associated with fund returns. This may seem to contradict the cost-efficiency argument. But it is important not to interpret this as evidence that fees themselves create returns. A more reasonable interpretation is that fees capture other differences between funds. Higher-fee funds may be different in several ways. They may have more active strategies, different sector exposure, more international exposure or a different investment style. The fee variable may therefore reflect not only the cost paid by investors, but also the type of strategy the fund follows. This is especially relevant because fees are strongly connected to the difference between active and passive funds.

The difference between gross and net returns is important for interpreting the fee result. Gross return shows how the fund performed before fees, while net return is closer to what investors actually receive. If funds with higher fees also have stronger gross returns, it may mean that these funds were exposed to strategies or assets that performed well during the period. The key question is whether this extra performance is large enough to compensate for the higher fees. The overall results suggest that this is not generally the case, since index funds still perform better than actively managed funds in the main models.

The fee interaction analysis adds further context. It suggests that the relationship between fees and returns is not the same for index funds and actively managed funds. For index funds, higher fees are harder to justify because the fund is mainly designed to follow an index, not to create extra returns through active decisions. For active funds, higher fees may be connected to a more differentiated strategy. The fund must then deliver enough additional return for the higher fee to be justified.

This is also connected to Cremers and Petajisto (2009). They argue that some active funds with a high active share may perform better, while funds that stay close to their benchmark are more likely to underperform after costs. This means that fees should not be analysed on their own. A higher fee may be easier to justify if the fund is genuinely active and clearly different from an index fund. If the fund behaves almost like an index fund, a high fee becomes more difficult to defend.

Therefore, the fee results should be interpreted carefully. The study shows that fees are associated with fund returns, but it does not show that higher fees are automatically good or bad. Instead, the result suggests that fees need to be understood together with fund type, investment strategy and whether returns are measured before or after fees.

In relation to the fee hypothesis, the results support an association between fees and returns. However, the positive coefficient suggests that the relationship may partly reflect fund characteristics associated with higher fees rather than fees themselves improving performance.

7.5 Standardised Risk Level and Fund Returns

According to Markowitz (1952), investors evaluate portfolios based on expected return and risk, and higher expected return is generally understood as compensation for bearing higher risk. Sharpe (1964) formalises this relationship in the Capital Asset Pricing Model, where systematic risk is expected to be rewarded through higher expected returns. Based on these theories, a positive association between risk and return could have been expected.

The standardised risk proxy does not show a statistically significant relationship with fund returns in the main models. This means that the study does not find evidence that funds with higher standardised risk proxy generated higher monthly returns during the analysed sample and period.

This may seem to conflict with Modern Portfolio Theory and CAPM, where higher risk is expected to be compensated by higher expected return. But the result should not be read as a rejection of these theories. The main reason is that the risk variable used in this thesis is not a direct market-based risk measure. It is not beta, volatility, downside risk or a Sharpe ratio. It is a standardised risk category presented to Swedish retail investors. This distinction matters. The standardised risk level is useful because it is comparable across funds and reflects the type of information that Swedish retail investors actually see when comparing funds. But it is a broad classification, not a precise estimate of financial risk. The variable is therefore useful as a control for general risk differences, but it is not strong enough to test the full theoretical risk-return relationship.

Another possible explanation for the results is that the risk variable may not vary enough across the sample. If most funds are concentrated around similar risk levels, the regression model has limited ability to identify a clear association.

A third point is that theory mainly discusses expected returns, while this thesis studies realised monthly returns over a five-year period. A higher-risk fund may have a higher expected return over the long run, but that does not mean it will generate higher realised returns in every specific five-year period. Market shocks, interest rates, sector movements and macroeconomic conditions can affect realised returns in ways that are not fully captured by the standardised risk proxy.

The negative but statistically insignificant risk coefficient should not be overinterpreted. Since the coefficient is not statistically significant, the interpretation is that standardised risk level does not explain monthly fund returns in this sample once the other variables are included.

In relation to the risk hypothesis, the result does not support the alternative hypothesis. The standardised risk level is not statistically associated with fund returns. This is still a useful finding, because it shows that the risk category investors see on fund platforms should not be treated as a direct return predictor.

7.6 Overall Analysis of the Findings

Overall, the results answer the research question by showing that sustainability grades and fees are associated with fund returns in the Swedish fund sample, while the standardised risk level is not statistically significant. The results also show that fund performance differs between index funds and actively managed funds, with index funds performing better.

The results are consistent with the implications of the Efficient Market Hypothesis and cost-efficiency arguments. Actively managed funds are expected to outperform the market through stock selection or market timing, but the empirical findings indicate that they do not outperform index funds in the aggregate sample. This is consistent with Fama (1970), Malkiel (1995), Carhart (1997), and Fama and French (2010), who all provide theoretical or empirical support for the argument that active managers often struggle to deliver persistent outperformance after accounting for risk factors and costs.

At the same time, the results should not be interpreted as a complete rejection of active management. Some actively managed funds may still outperform under specific conditions, during certain market cycles or through genuinely differentiated strategies. Cremers and Petajisto (2009) show that some funds with high active share may generate stronger performance, which suggests that the value of active management may depend on how genuinely active the fund actually is. The overall results in this thesis suggest that active management did not generate superior average performance in the Swedish fund sample during the analysed period.

The ESG result adds an important dimension. Stronger sustainability profiles are associated with higher returns, and the interaction model shows that this relationship is stronger among index funds than among actively managed funds. This means that ESG should not only be discussed as a separate sustainability variable. In this study, ESG is connected to performance, but its role depends on fund type. The finding is consistent

with Friede et.al (2015), who show that ESG is often positively or neutrally related to financial performance. The findings are also contrasted with the concern raised by Hong and Kacperczyk (2009), where sustainability-related exclusions may lead investors to miss exposure to high-return controversial assets. In this thesis, the empirical evidence is more consistent with the positive ESG interpretation than with the ESG constraint argument.

The fee result makes the analysis less straightforward. Cost-efficiency theory would suggest that higher fees reduce net returns, but the empirical results show a positive association in the main models. This should not be interpreted as proof that fees improve returns. A more careful interpretation is that fee levels may also capture broader differences in fund strategy and exposure. The fee robustness analysis supports this more careful interpretation, since the fee-return relationship appears to differ between active and passive funds.

The standardised risk level does not provide significant explanatory power in the models. This finding should be interpreted in light of the nature of the risk variable. Since the study uses a standardised fund risk category rather than a market-based risk measure, the result should not be viewed as a direct test of the risk-return relationship in Modern Portfolio Theory or CAPM. Instead, it suggests that the broad risk classification available to investors does not explain monthly fund returns in this sample once the other variables are included.

In conclusion, the findings suggest that Swedish fund returns in this sample are not explained by one single fund characteristic. Fund type, ESG Score and fees are relevant in explaining differences in returns in the sample, while the standardised risk proxy adds less explanatory value.

8. Conclusion

8.1 Conclusion

The purpose of this thesis was to examine whether fees and sustainability grades are associated with fund returns in Sweden when controlling for standardised fund risk level, and to analyse how performance differs between index funds and actively managed funds. Using monthly data from 84 Swedish funds over a five-year period, the thesis examines fund type, fees, sustainability grades and standardised risk level in relation to fund returns.

The findings show that fund type matters. In the main model specifications, actively managed funds are associated with lower returns compared to index funds. The finding is consistent with the implications of the Efficient Market Hypothesis, since the results do not indicate that active funds generate higher average returns than passive alternatives after costs. This suggests that it is difficult for active managers to consistently outperform the market through stock selection or market timing (Fama, 1970). It is also consistent with previous mutual fund research showing that actively managed funds often struggle to outperform passive alternatives after costs are taken into account (Malkiel, 1995; Carhart, 1997; Fama and French, 2010). In this study, the Swedish fund sample therefore provides support for the argument that index funds can offer stronger investor-relevant performance than actively managed funds.

The study also finds that sustainability grades are associated with fund returns. Since the Morningstar variable is a sustainability risk score, lower values indicate lower sustainability risk and therefore a stronger sustainability profile, the negative ESG coefficient indicates that funds with stronger sustainability profiles are associated with higher returns. The result should be interpreted as an association, not as proof that ESG ratings directly cause higher returns.

Fees are also associated with returns, but the result is not as simple as the theory would suggest. In theory, higher fees should reduce net returns because they are a direct cost to investors (Malkiel, 1995; Fama and French, 2010). In the empirical results in this study fees appear to capture more than just cost. They may also reflect differences in investment strategy, especially among actively managed funds. This means that fees should be interpreted together with fund type and gross versus net performance.

The standardised risk level does not show a significant relationship with fund returns. This does not mean that risk in general is unimportant, but rather that the risk classification used in this study is too broad to explain monthly returns in a precise way. The risk variable is best understood as a standardised investor-facing risk proxy, not as a full market-based risk measure.

The research question can therefore be answered as follows: fees and sustainability grades are associated with fund returns in the Swedish sample, while standardised risk level is not statistically significant. Fund performance also differs between index funds and actively managed funds, with the evidence generally favouring index funds during the analysed period.

8.2 Theoretical and Practical Implication and Contributions

8.2.1 Theoretical Implications

The theoretical implications of the study are mainly connected to market efficiency, fees, ESG and risk measurement. First, the results are consistent with the implications of the Efficient Market Hypothesis in the Swedish fund context. The fact that actively managed funds underperform index funds suggests that active managers, on average, did not create enough additional return to outperform passive alternatives after costs.

Second, the findings contribute to the discussion on cost efficiency. Fees matter, but the results show that they should not be interpreted too mechanically. In theory, fees reduce net returns. In practice, the fee variable may also reflect differences in fund strategy, market exposure and active management. This means that fees should be analysed together with fund type and the distinction between gross and net returns.

Third, the thesis contributes to sustainable finance literature by showing that ESG scores are statistically associated with fund returns in the Swedish sample. The result supports the view that sustainability characteristics can be financially relevant (Friede et al., 2015; Atz et al., 2023). At the same time, it does not remove the uncertainty in ESG research. ESG ratings may capture sustainability risk, but they may also be linked to other fund characteristics, such as sector allocation or investment strategy (Berg et al., 2022). This thesis provides context-specific evidence from Sweden, where stronger sustainability profiles are associated with higher returns during the analysed period.

The findings show that the way risk is measured matters. Modern Portfolio Theory and CAPM suggest that higher risk should be associated with higher expected return (Markowitz, 1952; Sharpe, 1964). The study contributes to the risk-return discussion by showing that the standardised risk level used in fund information did not have a statistically significant relationship with monthly fund returns in the main model. This does not reject the theoretical relationship between risk and return, but it suggests that broad investor-facing risk categories may be too general to explain short-term monthly return differences between funds. The finding therefore highlights a distinction between theoretical risk measures, such as beta or volatility, and standardised risk classifications used in retail fund information. This suggests that broad retail risk categories should not be treated as equivalent to beta, volatility or risk-adjusted performance. Theoretical discussions of risk need to distinguish between investor-facing classifications and more precise statistical risk measures.

8.2.2 Practical Implications

The practical implications are relevant. For private investors, the findings suggest that fund type and fees should be central when comparing funds. The results suggest that low-cost index funds may be attractive for investors seeking lower costs. This does not imply that all actively managed funds underperform, but it does mean that active funds need to justify their higher fees, which highlights the importance of fee justification for fund managers as well.

For investors who prioritize sustainable investing, the results are also relevant. Stronger sustainability profiles are associated with higher returns in the sample. This suggests that sustainability did not reduce performance during the analysed period. But, ESG

ratings should not be treated as a guarantee of future returns. They should be used together with other fund characteristics.

For fund platforms and financial advisors, the study shows the importance of explaining fund information clearly. Risk categories, ESG scores and fees are easy to compare, but they can also be misunderstood. A higher risk category should not be interpreted as a promise of higher return. A strong ESG score should not be interpreted as a guarantee of either ethical quality or future performance. Advisors and platforms therefore have an important role in helping investors understand what these measures actually mean.

The main contribution of this thesis is that it provides an integrated analysis of fund performance in the Swedish mutual fund market by combining four investor-relevant fund characteristics: fund type, fees, sustainability risk and standardised risk level. This thesis adds to the literature by examining these factors jointly and by separating index funds from actively managed funds. The findings therefore show not only whether fees and sustainability grades are associated with returns, but also how these associations appear when fund type and the standardised risk level communicated to investors are considered at the same time.

Overall, the practical message is that fund evaluation should not rely on one single number. Investors and other stakeholders should consider fund type, fees, ESG profile and risk classification together. The findings especially support careful evaluation of active management, attention to costs and a cautious but informed use of ESG information.

8.3 Ethical and Societal Implications

The ethical implications of this study are limited, since the thesis is based on secondary fund data and does not involve human participants, personal information or experimental treatment. The study therefore does not raise major ethical issues related to privacy, consent or direct harm.

The thesis still has societal relevance. Funds are a relevant part of household savings in Sweden, and many individuals use them for pensions, long-term savings and private investments (Finansinspektionen, 2025). Because of this, even relatively small differences in fees or returns can matter over time. A better understanding of how fees, sustainability grades, risk levels and fund type relate to performance can therefore support more informed investment decisions.

The sustainability dimension also gives the study broader relevance. ESG ratings are increasingly used by investors who want to combine financial goals with sustainability-related preferences. The results suggest that stronger sustainability profiles were not associated with lower returns in this sample. This is relevant because many investors may worry that sustainable investing requires a financial sacrifice. But, a fund with a strong ESG score should not automatically be seen as ethically superior and proof of future superior returns.

In summary, the ethical implications are not extensive, but the study has societal relevance because it concerns savings, sustainable investing and transparency in the Swedish fund market.

8.4 Limitations and future research

A limitation of this study is the measurement of risk. The study relies on the standardised risk classification collected through Avanza, which improves comparability and reflects the risk categories presented to Swedish retail investors. However, the measure is less precise than statistical measures such as beta, rolling volatility, downside risk or multifactor risk estimates. The variable should therefore be interpreted as a standardised categorical risk proxy for fund risk rather than a complete measure of financial risk exposure.

Future research could improve the risk measurement by estimating fund-specific volatility, beta, Sharpe ratios, downside-risk measures or multifactor model exposures. This would allow for a more detailed analysis of whether differences in fund returns remain after adjusting for market-based risk. A multifactor approach could also show whether fund performance is driven by systematic risk factors rather than by fund type, fees or ESG characteristics.

A second limitation is that several fund characteristics are not included in the empirical models. The study does not control for fund size, fund age, benchmark choice, fund flows, portfolio turnover, active share, sector allocation, geographic exposure or factor exposures such as value, growth, momentum and size. These factors may influence fund returns and may also be related to fees or ESG scores. For example, funds with stronger sustainability profiles may have different sector exposures than funds with weaker sustainability profiles. This could partly explain the ESG-return relationship. Future research should include more detailed fund-level controls to examine whether the results remain robust.

The fact that the study uses one provider of the sustainability risk score can be a limitation. The measure is relevant because it is visible to investors and widely used in fund evaluation. But, ESG ratings differ between providers because they use different methods, data sources and weightings (Berg et al., 2022). Future research could compare ESG ratings from different providers or analyse the environmental, social and governance dimensions separately.

Another opportunity for future research related to the ESG scores is that these are not fixed. Due to limited access to historical Morningstar ESG data, each fund was assigned the ESG score available on 10 April 2026. This limits the ability to capture changes in sustainability risk over the sample period.

Another limitation is the five-year time period. The period is useful because it captures recent fund performance and improves comparability of ESG data. At the same time, five years is still a limited period in financial markets. The results may be affected by specific market conditions, such as interest rate changes, sector rotations, currency movements or macroeconomic shocks. Future research could extend the time period or divide it into sub-periods to test whether the relationships are stable over time.

The focus on the Swedish context also acts as a limitation for the study. The delimited regional context makes the study relevant for Swedish investors, but it also limits how far the findings can be generalised. Future research could compare Sweden with other Nordic or European markets to see whether the same relationships exist elsewhere.

Further limitations are the classification of funds as either index funds or actively managed funds. This distinction is central to the thesis, but in practice the boundary between passive and active management is not always binary. Some funds may use enhanced indexing, semi-active strategies or portfolios that stay close to their benchmark. Future research could include active share or tracking error to measure how active the funds actually are.

In conclusion, these limitations do not invalidate the findings, but they define the conditions under which the results should be interpreted. Future research can build on this thesis by using more precise risk measures, additional fund-level controls, alternative ESG data, longer time periods and more detailed measures of active management.

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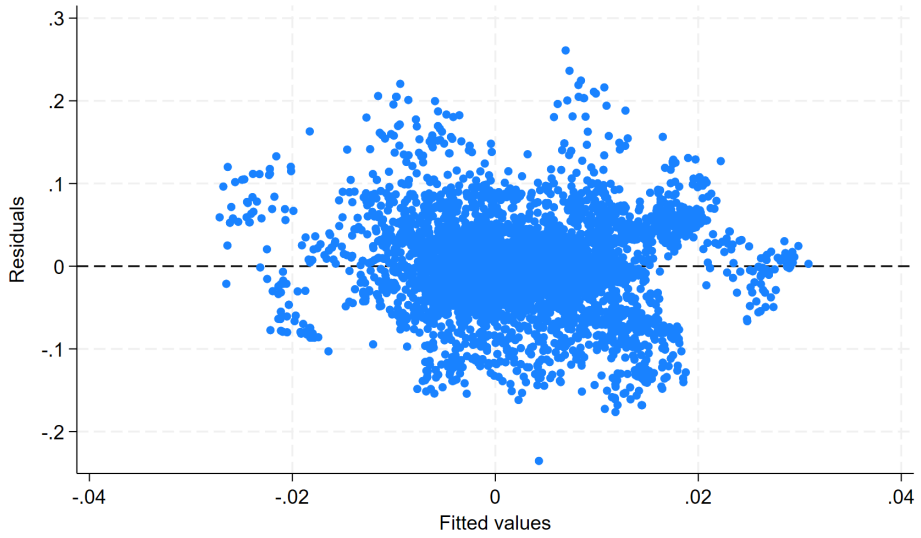
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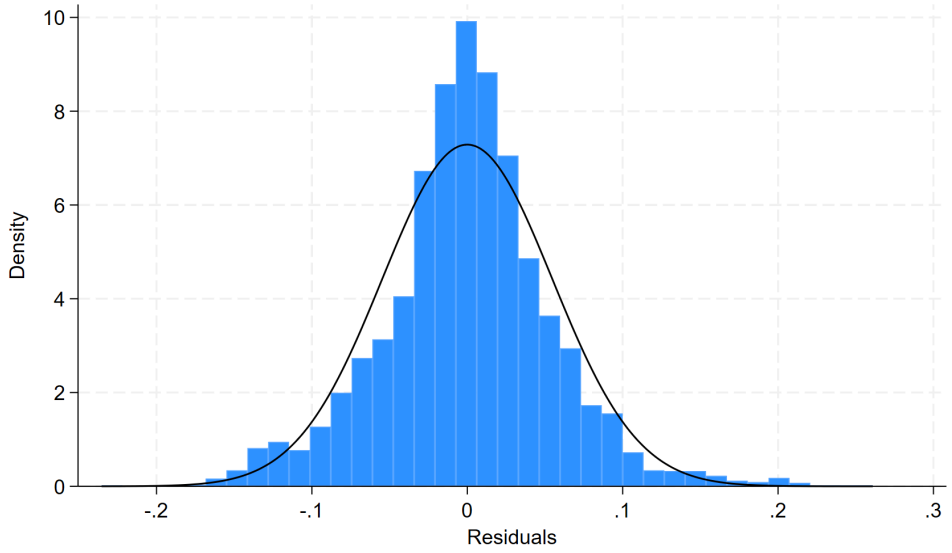
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Appendix

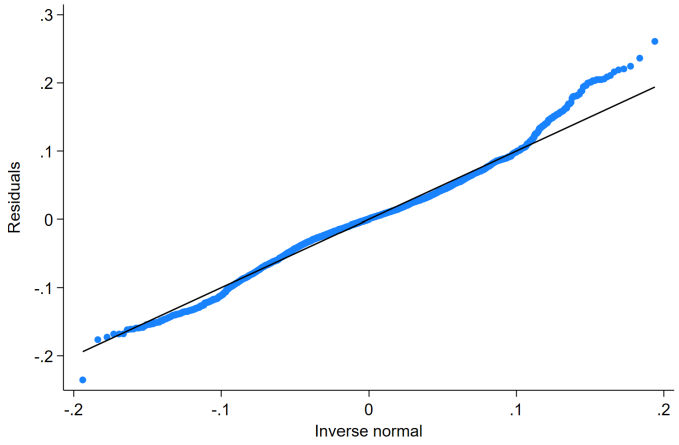
Appendix 1: Residual-versus-fitted plot



Appendix 2: Histogram of residuals



Appendix 3: Q-Q plot of residuals



Appendix 4: Breusch-Pagan test output

```
Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: Normal error terms
Variable: Fitted values of net_return

H0: Constant variance

    chi2(1) =   0.07
Prob > chi2 = 0.7962
```

Appendix 5: Fund list

NAMN	ID	Index 1 /Aktiv 2
Avanza Zero	1	1
Aktiespararna Topp Sverige A	2	1
Aktiespararna Direktavkastning A	3	1
Handelsbanken Sverige 100 Index	4	1
Lannebo/Öhman Marknad Sverige A	5	1
Aktiespararna Direktavkastning B	6	1

Nordea Sverige Passiv icke-utd	7	1
PLUS Allabolag Sverige Index	8	1
Storebrand Sverige B SEK	9	1
Storebrand Sverige A SEK	10	1
Danske Invest Sverige Beta SA	11	1
Swedbank Robur Access Sverige	12	1
SEB Sverige Indexnära B utd	13	1
SEB Sverige Indexnära A	14	1
Skandia Sverige Exponering	15	1
Handelsbanken Sverige Index Criteria	16	1
Swedbank Robur Access Edge Sverige A	17	1
SEB Sverige Indexfond B	18	1
Storebrand Sverige Plus B SEK	19	1
Storebrand Sverige Plus A SEK	20	1
PLUS Småbolag Sverige Index	21	1
Storebrand Sverige Småbolag Plus B SEK	22	1
Storebrand Sverige Småbolag Plus A SEK	23	1
PLUS Mikrobolag Sverige Index	24	1
SEB Swedish Value Fund	25	2
SEB Sverigefond	26	2
Lannebo Sverige Plus E	27	2
Lannebo Sverige A SEK	28	2
Handelsbanken Sverige Selektiv	29	2
Kvartil Investmentbolag+Calculus A	30	2
Länsförsäkringar Sverige Vision A	31	2
Carnegie Sverigefond A	32	2

PriorNilsson Sverige Aktiv A	33	2
Carnegie Småbolagsfond A	34	2
Indecap Sverige C	35	2
PriorNilsson Evolve A	36	2
SEB Sverige Expanderad	37	2
Nordea Ideell Aktiefond	38	2
SEB Sweden Equity C (SEK)	39	2
Simplicity Sverige	40	2
Alcur Grow B	41	2
Case Högutdelande Bolag A	42	2
Cliens Sverige B	43	2
Cicero Sverige A	44	2
Aktie-Ansvar Sverige A	45	2
AMF Aktiefond Sverige	46	2
Danske Invest Sverige SA	47	2
Ohman Sverige Fokus A	48	2
Humle Sverigefond	49	2
Swedbank Robur Sverige A	50	2
Carnegie All Cap A	51	2
Spiltan Aktiefond Småland	52	2
SEB Sverigefond Småbolag	53	2
C WorldWide Sweden 1A	54	2
Spiltan Aktiefond Stabil	55	2
Cliens Småbolag A	56	2
SEB Sverigefond Småbolag C/R	57	2
Nordea Småbolagsfond Sverige	58	2

Cliens Sverige Fokus A	59	2
Skandia Cancerfonden	60	2
Pareto Sverige A	61	2
Lannebo Småbolag A	62	2
D&G Aktiefond	63	2
Humle Småbolagsfond A	64	2
Simplicity Småbolag Sverige A	65	2
Simplicity Småbolag Sverige B	66	2
Danske Invest Sverige Småbolag SA SEK	67	2
Lancelot Sverige B	68	2
ODIN Small Cap C SEK	69	2
Handelsbanken Sverige (A1 SEK)	70	2
Cliens Micro Cap A	71	2
Lancelot Sverige A	72	2
Consensus Sverige Select A	73	2
C Worldwide Sweden Sall Cap 1A	74	2
Swedbank Robur Småbolagsfond Sverige A	75	2
Spiltan Småbolagsfond	76	2
Avanza Småbolag by Skoglund	77	2
D&G Småbolag	78	2
Skandia Småbolag Sverige	79	2
Pareto Småbolagsfond A	80	2
Länsförsäkringar Småbolag Sverige Vis A	81	2
Indecap Guide Q30 C	82	2
AMF Aktiefond Småbolag	83	2
Handelsbanken Svenska Småbolag (A1 SEK)	84	2

Appendix 6: Summary of variables used in previous research.

Study	**Sample/ context**	**Dependent variable**	**Main independent variables**	**Control variables**	**Management type examined**	**Main finding**
Malkiel (1995)	US mutual funds	Fund performance/ returns	Fees and expenses	Fund characteristics and market benchmarks	Active and passive funds	Actively managed funds generally underperform passive alternatives after costs
Carhart (1997)	US mutual funds	Risk-adjusted fund performance	Expense ratios, turnover and load fees	Market, size, value and momentum factors	Primarily actively managed funds	Costs are negatively associated with fund performance
Fama and French (2010)	US mutual funds	Abnormal fund returns	Management performance and costs	Common risk factors	Actively managed funds	Limited evidence of persistent managerial skill after costs
Cremers and Petajisto (2009)	US equity funds	Benchmark-adjusted returns	Active share and expenses	Benchmark exposure and fund characteristics	Actively managed funds	Highly active funds may outperform, while low-active-share funds often underperform
Current study	Swedish mutual funds	Fund returns	Fees and Morningstar sustainability measure	Standardised fund risk level	Index and actively managed funds	Examines whether the associations differ between management types



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